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Advancements and prospects of OCT-enabled all-process monitoring and inline quality assurance in laser keyhole welding: A critical review

Di Wu ^{a b}  , Peilei Zhang ^{a b}, Haichuan Shi ^{a b}, Qinghua Lu ^{a b}, Zhishui Yu ^{a b}, Yuze Huang ^c [Show more](#) ✓[Download full issue](#)[Cite](#)[Add to Mendeley](#)[Share](#)[10.1016/j.jmapro.2025.08.063](#) ↗

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Abstract

Laser keyhole welding (LKW) is widely used in manufacturing, yet its inherent instabilities limit its application in high-value product production with strict requirements. An *in-situ* process monitoring system is thus essential to overcome this hurdle. Optical coherence tomography (OCT), a high-precision laser ranging technique, can directly measure keyhole depth with micron-scale resolution within microseconds. Consequently, our review commences by delineating the ongoing research efforts focused on OCT-based monitoring and inline quality assurance (QA) during the dynamic LKW process. Subsequently, we conduct a meticulous analysis of the operating principles of

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diverse OCT techniques, along with an examination of OCT-data processing and precision analysis. In particular, we summarize the various categories of all-process monitoring and their corresponding applications. This endeavor aims to deepen our fundamental understanding of the process-signature-quality relationship. Furthermore, we explore the latest advancements in other state-of-the-art sensing and machine learning (ML) techniques and investigate how these can be integrated into the OCT measurement system. Finally, we engage in a discussion regarding the major challenges that exist in three aspects and propose potential solutions. Overall, this comprehensive review paves the way for more in-depth investigations into the underlying physics of the LKW process, aiding deterministic modeling and real-time process control.

Keywords

Laser keyhole welding; Optical coherence tomography; Data processing; Machine learning; Quality assurance

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1. Introduction

1.1. Physics of laser-metal interaction

As an advanced material joining method, laser keyhole welding (LKW) can enable a high-intensity laser source dramatically evaporate the metal workpiece to a certain temperature, and drills a deep and narrow vapor capillary (called keyhole) inside the melt pool [1]. By virtue of a great many advantages of big penetration depth, high speed welding, low heat input, small heat deformation, and ease of precise automatic control, LKW can ensure the sufficient penetration at travel speed higher than traditional joining techniques. Thus it significantly improves the welding efficiency and product quality,

6. Combination of OCT with machine learning

7. Major challenges and potential solutions

8. Conclusions

CRediT authorship contribution statement

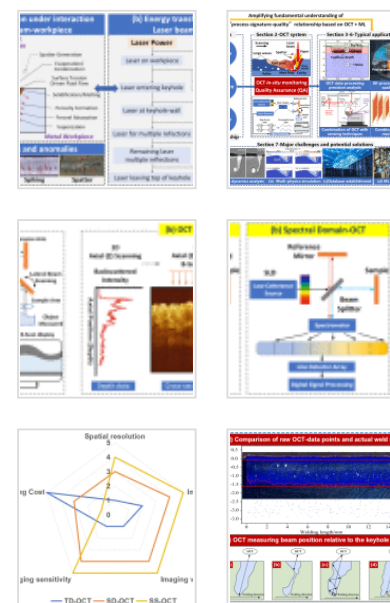
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References

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Figures (16)



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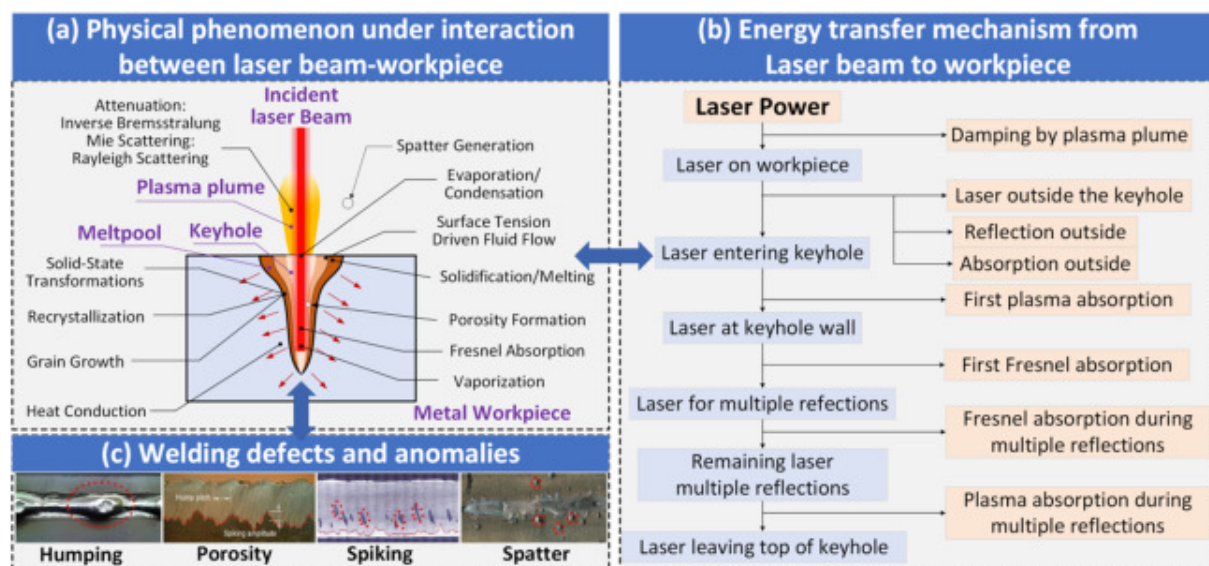
which is widely applied in many manufacturing applications *e.g.* defense systems, aerospace, medical and automotive industry [2].

As depicted in Fig. 1(a), within the context of high-irradiance laser-metal interactions, a vapor (keyhole) channel, reaching depths of up to several millimeters, is induced in the region traversed by the process beam. This keyhole is encompassed by a molten metal volume that, upon cooling, gives rise to the weld seam. Concurrently, the dynamic keyhole maintains its openness under the equilibrium of laser-induced evaporation recoil pressure, surface tension, and hydrostatic pressure [3]. Notably, the keyhole exhibits a slender profile, tapering mildly toward the base and curving in the trailing direction. Such curvature plays a pivotal role in modulating the light-matter interaction effect. To precisely simulate the keyhole morphology, due consideration must be given to the laser energy coupling dynamics throughout the laser welding (LKW) process. Fig. 1(b) elucidates the energy transduction from the laser source to the keyhole walls, mediated by diverse interaction mechanisms. The multiplicity of reactions constitutes a crucial determinant for keyhole formation and stability, enabling the efficient absorption of the preponderant portion of the incident power at specific locations within the keyhole. Indeed, the keyhole is perceived as a tortuous entity far from dynamic stationary, which has numerous properties and features commonly associated with turbulence behavior [4]. Consequently, even under routine operational conditions, a wide variety of defects and anomalies tend to emerge with relative ease, as depicted in Fig. 1(c). The prevalent defects and anomalies principally comprise: 1) ripples on the front wall (humping), which manifest as periodic undulations on the leading edge of the weld pool; 2) irregular depth oscillation (spiking), characterized by unpredictable fluctuations in the vertical dimension of the keyhole; 3) keyhole-induced collapse (porosity); and 4) ejection of liquid metal (spatter), the molten material is forcefully expelled from the weld zone, potentially contaminating the surrounding area and degrading weld quality.

Table 1. Comparison of resolution and i...

Table 3. Summary of different OCT data ...

Table 3. Summary of combination of OC...



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Fig. 1. Schematic showing physical dynamics during LKW process: (a) Physical characteristics under a high-irradiance laser-metal interaction; (b) Energy transfer mechanism from laser to workpiece based on the plasma and Fresnel absorption effects [4]. (c) A wide variety of welding defects and anomalies.

1.2. The context of OCT monitoring

It is widely acknowledged that the aforesaid potential defects exert a pronounced negative impact, substantially undermining the mechanical properties and heightening the fatigue susceptibility of welded components. This, in turn, gravely compromises the service life expectancy of the product. A principal hurdle in the expansion of laser processing across diverse industrial applications lies in the holistic guarantee of weld seam quality. Especially in performance-sensitive applications, the certification of parts fabricated *via* laser welding persists as a formidable impediment [5]. Theoretical analysis and experiments indicate that the keyhole hardly remains in stable status, even in the

optimal process space. Current analytic and numerical modeling approaches have been applied to characterize and model the complex physics system for decades, yet many adopted hypotheses may cause unacceptable discrepancies between the computed and actual welding results. Moreover, such established self-consistent numerical models are theoretically sophisticated and computationally expensive, and thus cannot be applied for immediate consulting [6].

In this regard, to ensure the production of welds free from defects and to meet the diverse quality criteria, even when confronted with process fluctuations and manufacturing tolerances, *in-situ* monitoring is vital and fundamental for ensuring continuous traceability and shortening development cycles in the sense of broad quality assurance. According to the summary of several published reviews [[7], [8], [9], [10], [11]], there are four representative monitoring techniques to measure aspects of the keyhole behavior. It mainly includes the coaxial/paraxial visual [[12], [13], [14]], optical [[15], [16], [17]], acoustic [[18], [19], [20], [21]], and temperature detection [22,23]. Recent efforts have made great progress in extracting the essential features from acquired multi-source signals, and building the corresponding relationship between them and keyhole dynamics, thus further predicting/estimating the resultant welding quality. Nevertheless, the laser welding (LKW) process is currently hampered by a scarcity of experimental methodologies that permit direct observation in the direction of the beam and a comprehensive analysis of keyhole dynamics. Instead of relying on the correlation of indirect measurements with theoretically predicted outcomes, the pursuit of accurately gauging keyhole behavior has proven to be an elusive goal. To date, X-ray transmission videography [24] stands as the most straightforward measurement technique for keyhole geometry and dynamics. It enables precise quantification of the internal keyhole profile hidden beneath the metal workpiece. However, its application is markedly constrained by the stringent environmental prerequisites and the restricted accessibility of synchrotron light sources. Also, these are expensive experiments to carry out in general laboratory environment or industrial site.

In the realm of process monitoring, Optical Coherence Tomography (OCT) has emerged as a rapidly developing and highly promising imaging tool. This is due to a host of remarkable advantages it offers, including non-contact measurement, high measurement accuracy and high imaging speed [25]. With the lack of a powerful X-ray source in many scenarios, an increasing number of researchers have embarked on exploring the potential and practical applicability of OCT within the domain of laser welding. Rooted in the principle of coherence interferometry-based emerging imaging, OCT can be seamlessly integrated coaxially with the laser beam. This integration transforms it into an invaluable asset for on-site, *in-situ* monitoring and control of laser welding quality. As a result, numerous studies have centered around OCT's capabilities, such as its proficiency in estimating seam depth with great accuracy, its effectiveness in tracking the seam online throughout the welding process, and its capacity to identify weld quality issues promptly [[26], [27], [28]]. In fact, a few OCT systems have already found their way into the commercial market, particularly in the electronics and automobile industries, where the demand for high-quality welds is ever-present [29,30].

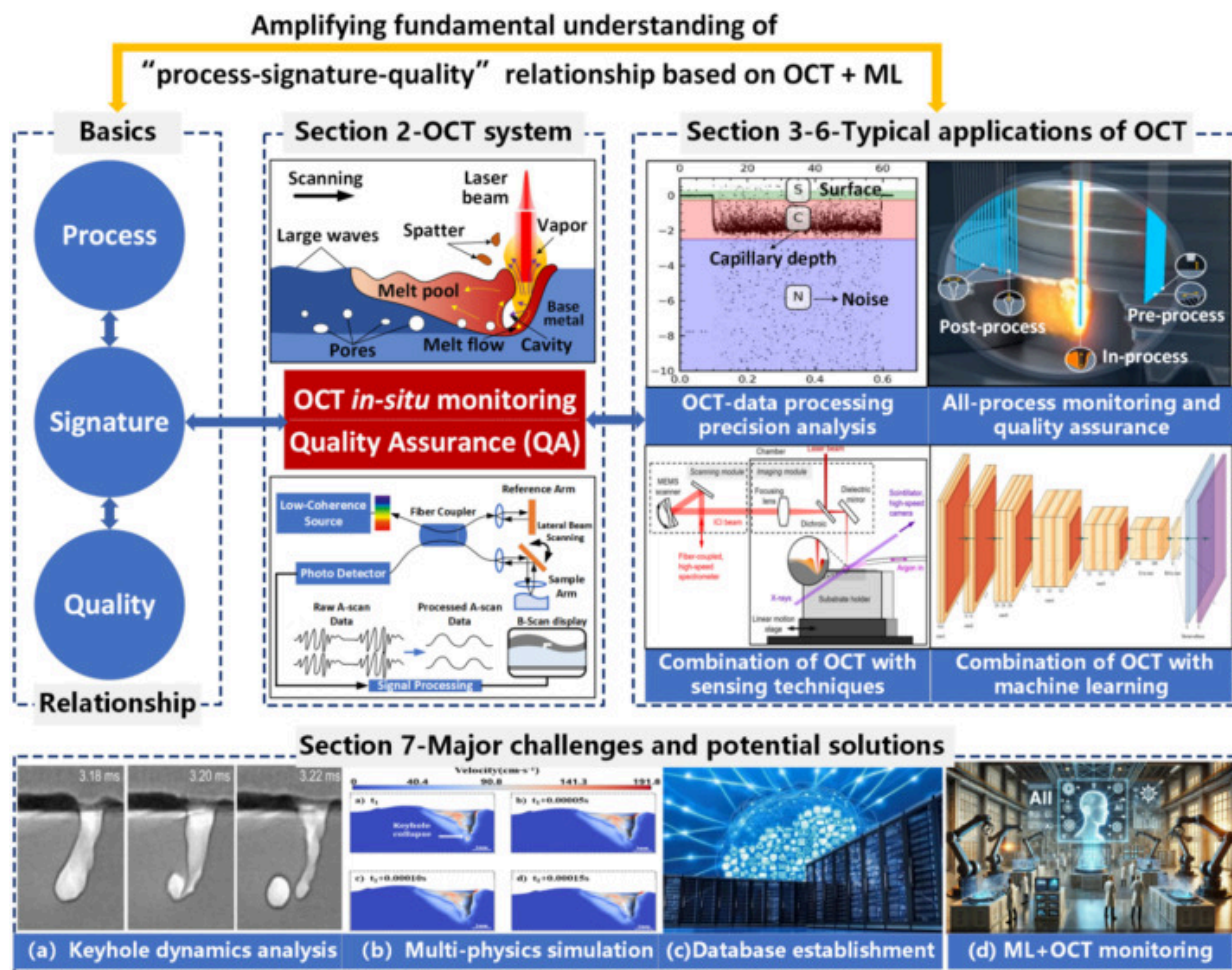
1.3. Major contributions of our review

In order to augment the in-depth understanding of the intricate mechanisms and latent defects inherent in the LKW process, it is imperative to conduct a meticulous exploration of the causal relationships interconnecting the process parameters, process signatures, and welding qualities. As elaborated in our published review [10], the process parameters associated with Key-Input-Characteristics (**KICs**) serve as the pivotal “model inputs”, exerting a preponderant influence in dictating the precise quantum of laser energy delivered to the workpiece. With respect to the process signatures, within the purview of this review, those linked to Key-Measurement-Characteristics (**KMCs**) predominantly consist of OCT-based signals that emanate from the dynamic fluctuations of the keyhole during the laser-metal interaction. When contemplating the welding quality, which is defined in terms of Key-Performance-Characteristics (**KPCs**), welding defects, penetration depth, and seam width emerge as the cardinal factors that directly govern the weld quality.

Overall, there exists an urgent necessity for us to achieve a comprehensive apprehension of the “process-signature-quality” nexus, thereby endowing us with the capacity to accurately delineate this complex physical system. Apparently, the OCT technology has the potential to decipher a great number of intricate causal mysteries. However, the precise interpretation and processing of its data present formidable challenges and are far from being facile. Therefore, the exigency for a comprehensive and profound exploration and dissection of OCT measurement constitutes the principal driving force behind the compilation of this review. It accentuates the potential of synergistically integrating OCT with laser keyhole welding to actualize real-time defect rectification. To date, in comparison with the previously published reviews and papers, the major contributions of our work are concluded as follows:

- (1) To the best of our current academic knowledge at present, this review represents a pioneering endeavor as it proffers a systematically organized summary dedicated to OCT monitoring and inline quality assurance throughout the LKW process.
- (2) Our study presents an elaborate and detailed summary, with the primary objective of introducing a diverse array of OCT-data processing methodologies. In addition, it delves deep into unveiling the intricate correlation subsisting between the influencing factors and OCT-data, which can facilitate a robust and reliable interpretation of the OCT results.
- (3) Our review comprehensively encapsulates the latest advancements in all-process monitoring, achieved through the synergistic integration of various sensing and ML approaches. This holistic overview further aids readers in enhancing their comprehension of the complex laser-metal interactions, which proffers profound insights into the underlying physics of laser welding.

The organizational framework of this review is visually represented in Fig. 2. initiates with a comprehensive and detailed exposition regarding the underlying physics of laser-metal interaction, as well as the monitoring context prevalent during the LKW process. Section 2 undertakes the task of aggregating and contrasting the fundamental principles and performance evaluation criteria of diverse OCT techniques. Section 3 conducts a methodical and systematic review of the cutting-edge OCT-data processing methodologies. It places particular emphasis on dissecting the multiple factors that play a crucial role in OCT precision analysis. Section 4 compiles a summary of all-process monitoring modalities and their associated techniques. Subsequently, Section 5 and Section 6 jointly offer an in-depth assessment of the OCT monitoring in conjunction with other advanced sensing modalities and ML approaches. Finally, Section 7 delineates the potential prospects and existing challenges associated with the OCT monitoring of the laser welding process. Section 8 then culminates this extensive review.



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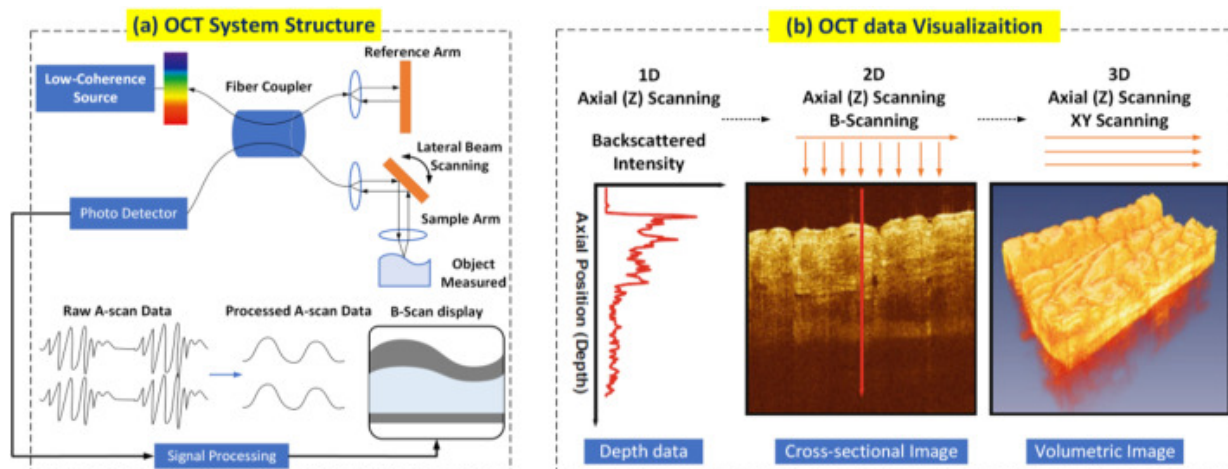
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Fig. 2. A flowchart of OCT-based all-process monitoring and inline QA throughout the LKW process, which encompasses the following key sections: Section 2-OCT principle and structure; Section 3-OCT-data processing and precision; Section 4-All-process monitoring and quality assurance; Section 5-Combination of OCT with sensing techniques; Section 6-Combination of OCT with ML; Section 7-Major challenges and potential solutions.

2. Operation principle and performance of OCT

2.1. Structure and data visualization of OCT

In 1991, David Huang and his colleagues [25] from the Massachusetts Institute of Technology made a pioneering breakthrough by inventing and subsequently publishing their research findings regarding OCT technique. It has since emerged as a rapidly advancing optical imaging modality, finding extensive applications in the fields of biomedicine and ophthalmology. As depicted in Fig. 3(a), a near-infrared laser beam originating from a low-coherence light source is directed through a 2×2 fiber-optic Michelson-type interferometer and then proceeds to impinge upon a sample. It is assumed that the coupler divides the incident laser power equally between the sample arm and the reference arm. The backscattered light intensity from the sample is then re-coupled into the interferometer. Subsequently, the optical path length information becomes encoded within the spectral interference pattern. Consequently, OCT stands out as a powerful imaging technique. Its prowess lies in its ability to conduct cross-sectional imaging (referred to as “Optical Biopsy”), which is achieved by *in-situ* measurement of the magnitude and echo time delay of the backscattered light [31,32]. This unique capability allows for detailed visualization of internal structures within biological tissues or other relevant samples without the need for invasive procedures, thereby making it an invaluable tool in both research and clinical settings.



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Fig. 3. Structure and data visualization of OCT measurement.

As illustrated in Fig. 3(b), OCT represents a state-of-the-art 3D tomography technology predicated on the imaging principle of low-coherence interferometry (LCI). Essentially, it functions by meticulously measuring the interference signals between backscattered light and reference light, thereby extracting the structural details in the depth dimension of the sample, a process known as axial scans or A-scan. Subsequently, through transverse scanning, a two-dimensional structural representation (B-Scan) of the sample can be obtained. This process culminates in the generation of a two-dimensional data set that faithfully mirrors the optical backscattering occurring within a cross-sectional plane traversing the tissue. Such data can be presented either in false color or grey scale, facilitating the visualization of tissue pathology and enabling clinicians and researchers to discern intricate details about the sample's internal structure. Finally, by systematically acquiring sequential cross-sectional images *via* raster pattern scanning of the incident optical beam, comprehensive three-dimensional OCT (3D-OCT) data is produced. This volumetric data encapsulates comprehensive structural information and exhibits operational similarities to computed tomography (CT) or magnetic resonance (MR)

images, making it a convenient and powerful tool for a variety of applications. It allows for easy comparison and integration with existing imaging modalities, further enhancing its utility in the medical and scientific communities [33].

To gain a more comprehensive and profound understanding of the OCT imaging technique, it is highly instructive to draw comparisons with the commonly employed ultrasound and microscopy techniques, as meticulously detailed in Table 1. The resolution of standard clinical ultrasound imaging typically spans from 0.1 to 1 mm and is.

Table 1. Comparison of resolution and imaging depth for different OCT techniques.

Imaging technique	Imaging resolution	Penetration depth	Advantages and drawbacks
Ultrasound	0.1–1 mm	4–8 mm	Advantage: with the advent of high-performance and low-cost digital signal processing technology, high-frequency ultrasound now possesses a more refined resolution;
	15~20 μm	1–2 mm	Drawback: strongly attenuated in sample with a limited imaging depth.
Confocal microscopy	1 μm	200~500 μm	Advantage: having an extremely high resolution (about 1 μm); Drawback: the imaging depth is constrained by optical scattering.
OCT	1~15 μm	1–5 mm	Advantage: high resolution of OCT imaging enables the rapid and accurate visualization of sample inside morphology. Drawback: most samples cause high scattering of OCT light, the attenuation due to scattering restricts the image penetration depth to approximately 5 mm.

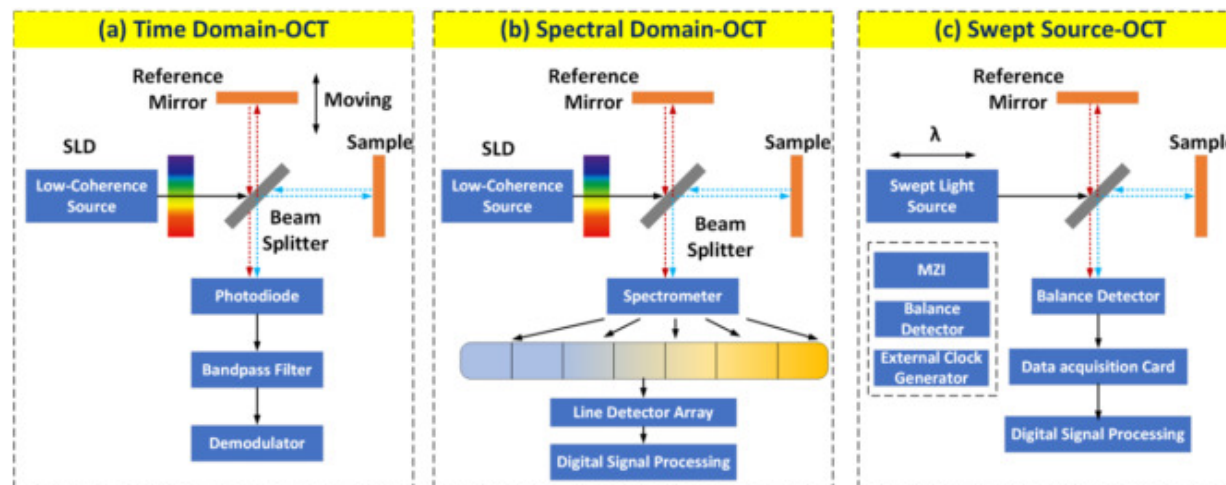
intrinsically linked to the sound wave frequency (ranging from 3 to 40MHz) harnessed for the imaging process [34]. Notably, when a higher ultrasound frequency (approximately 100MHz) is utilized, it can yield a remarkably finer resolution, down to 15–20 μ m. However, this enhanced resolution comes at a cost, as the imaging depth is severely curtailed to just a few millimeters due to the pronounced attenuation that occurs within the sample. In stark contrast to ultrasound imaging, confocal microscopy flaunts an extraordinarily high resolution, approximately 1 μ m, which is fundamentally dictated by the light diffraction limit. Regrettably, its imaging depth within the sample is confined to merely a few hundred micrometers. This limitation arises because the optical scattering effect precipitates significant deteriorations in both the image signal and contrast.

OCT artfully bridges the gap that exists between ultrasound, renowned for its millimeter-level imaging depth, and microscopy, celebrated for its micrometer-level resolution. The axial image resolution in OCT is predominantly governed by the light source bandwidth. Presently, existing OCT technologies offer axial resolutions varying from 1 to 15 μ m, which represents an almost 10–100-fold improvement over standard ultrasound imaging. This high-resolution prowess of OCT imaging empowers the visualization of the internal morphology of samples with remarkable clarity. In recent times, OCT has ascended to become a clinical standard in ophthalmology, given that the eye can be non-invasively imaged with great facility and fine resolution [35]. Nevertheless, the OCT technique is not without its drawbacks. A principal disadvantage is that light undergoes high levels of scattering when interacting with most samples. This scattering-induced attenuation restricts the image penetration depths to approximately 5mm. However, a particularly outstanding characteristic of OCT lies in its remarkable versatility as an optical modality. It demonstrates seamless compatibility and can be effortlessly incorporated with a wide variety of medical instruments, including endoscopes, catheters, laparoscopes, and needles. This integration paves the way for effective internal body imaging, expanding the scope of its clinical applications. In the realm of laser welding monitoring, the OCT probe detector can be effortlessly combined with custom-made laser process heads, enabling the *in-situ* and real-time acquisition of keyhole depth information. In the following

section, we will embark on a comprehensive review of the physical fundamentals underlying OCT techniques.

2.2. Fundamentals of OCT in time- and Fourier-domains

Based on the nature of the detected signals, OCT can be categorized into two main types: TD-OCT (Time-domain OCT) and FD-OCT (Fourier-domain OCT). Specifically, by dissociating the wavelength from the spatial domain and the wavelength from the time domain, the power spectral density of the interference signal can be acquired. Leveraging this principle, FD-OCT can be further subdivided into two subtypes: SD-OCT (Spectral-domain OCT) and SS-OCT (Swept-source OCT). Fig. 4 illustrates the fundamental aspects of these three distinct OCT techniques, providing a visual representation that helps in understanding their unique characteristics and operational differences.



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Fig. 4. Comparison between three variants of different OCT techniques: TD-OCT, SD-OCT and SS-OCT.

2.2.1. TD-OCT technique

As depicted in Fig. 4(a), TD-OCT technique primarily consists of several key components, namely a near-infrared low-coherence light source, a Michelson interferometer, and a photodiode detector. The light emitted from a broadband superluminescent diode (SLD) undergoes a splitting process after passing through the fiber coupler. As a result, it is divided into two sub-beams that possess identical frequency, a constant phase difference, and the same vibration direction. Subsequently, these two sub-beams follow different paths: One of the sub-beams travels toward the reference mirror, reflects off it, and then returns to the interferometer. This sub-beam is referred to as the reference light. The other sub-beam simultaneously impinges on the sample, gets reflected or scattered by the sample, and then returns back to the interferometer. This is known as the sample light. Both of these sub-beams then interfere with each other within the fiber coupler. This interference event generates interference information that is detected by the photodiode detector. Consequently, the TD-OCT technique acquires the interference information through longitudinal depth scanning, which is based on the signal delay and intensity. By means of this process, it becomes possible to extract the reflectivity information corresponding to the sample's depth. When combined with the results obtained from transverse scanning, this enables the achievement of tomography, allowing for a comprehensive visualization of the internal structure of the sample [36].

Suppose there exists a reflection point along the optical axis direction of the sample light. In this case, the intensity of the interference signal measured by the photodetector can be expressed as follows:

$$I(\Delta l) = R_R^2 I_0 + R_S^2 I_0 + 2R_R R_S \operatorname{Re}\left\{\Gamma_0\left(\frac{n\Delta l}{c}\right)\right\} \quad (1)$$

Here, Δl represents the optical path difference between the reference mirror and the sample reflection point d , R_R and R_S denote the reflection coefficients of the reference mirror and the reflection point d , respectively. I_0 is the light intensity incident on the reference arm and the reflection point d . Γ_0 is the mutual coherence function of the reference light and the sample light, and n is the refractive index. It should be noted that the first and second terms on the right-hand side of formula (1) correspond to the self-

coherent light intensities of the reference light and the sample light, respectively. These terms belong to DC noise. In addition, the third term represents the mutual coherence function, which is directly associated with the depth information of the reflection point d .

Due to the fact that the coherence length of the broadband light source is relatively short, an interference fringe can be detected only when the optical path difference Δl between the reference light and the sample light is less than the coherence length. This situation corresponds to the third term in [formula \(1\)](#). Consequently, by measuring the interference intensity, it is possible to obtain the depth information of the sample and the reference mirror with an equivalent optical path. For multiple reflection points along the optical axis direction of the sample, the depth information in the direction of the sample's optical axis can be extracted through the following process: the reference mirror is moved for depth scanning, and during this process, the optical path of the reference mirror as well as the corresponding interference signal intensity are recorded. Subsequently, the obtained depth information can be integrated. By changing the optical axis of the sample arm for transverse scanning, the tomography of the sample can be realized [\[37\]](#).

Consequently, TD-OCT boasts the advantage of point detection. The measurement depth in this technique is determined by the scanning range of the reference mirror. Nevertheless, TD-OCT is fraught with several significant shortcomings. It suffers from a relatively long scanning time, which leads to a slow imaging speed. Additionally, it exhibits high levels of noise and low sensitivity. These drawbacks are primarily attributable to the physical limitations that exert an impact on multiple aspects of the TD-OCT technique, including measurement, time consumption, sensitivity, and resolution. Such limitations can impede its performance in various applications where rapid and accurate imaging with high sensitivity is required.

2.2.2. FD-OCT technique

To more effectively tackle the aforementioned issues, a growing number of scientific studies have opted to employ Fourier-domain-based OCT (FD-OCT). Compared to the

traditional Time-domain OCT (TD-OCT), FD-OCT can significantly enhance the acquisition rate. Specifically, FD-OCT has the capacity to acquire the mutual coherence function of interference signals. By conducting an inverse Fourier transform on the cross-power spectral density of these interference signals, it can extract the depth information of the sample. Through this approach, FD-OCT can circumvent the time-consuming depth scanning process. As a result, it can improve the imaging speed and sensitivity while simultaneously reducing the scanning noise. Owing to these advantages, FD-OCT has emerged as a mainstream OCT technology.

In 1995, Fercher et al. [38] were the first to propose the Spectral-domain OCT (SD-OCT) technique. SD-OCT mainly consists of three components: a near-infrared broadband light source, a Michelson interferometer, and a spectrometer, as illustrated in Fig. 4(b). With the utilization of an optical grating, SD-OCT separates the interference signal from the space domain based on wavelength. Subsequently, the interference light of different wavelengths is focused onto different pixel units of a Charge-Coupled Device (CCD) through a convergent lens. This process enables the acquisition of the cross-power spectral density of the interference signal. Furthermore, Yasushi, et al. [39] put forward a Swept-source OCT (SS-OCT) technique, which comprises a broadband swept laser source, a Michelson interferometer, and a photodiode balanced receiver acting as the detector, as illustrated in Fig. 4(c). In the SS-OCT system, the optical wavelength emitted by the swept laser source is isolated from the time-domain. Subsequently, the returned interference signal can be gathered sequentially by the photodetector over time, facilitating the acquisition of the cross-power spectral density.

Suppose that the optical path of the reference light is denoted by l_R , and the reflection coefficient of the reference mirror is represented as R_R . Moreover, suppose that there are n reflection points along the optical axis direction of the sample light. The corresponding optical path for each of these reflection points is l_i , and the reflection coefficient is R_i (where $i=1,2, \dots,n$). Under these assumptions, the cross-power spectral density of the

interference signal can be obtained as follows [39]:

$$S(k) = R_R^2 S_0(k) + \sum_{i=1}^n R_i^2 S_0(k) + \sum_{i \neq j}^n 2R_i R_j S_0(k) \cos[k(l_i - l_j)] + \sum_{i=1}^n 2R_R R_i S_0(k) \cos[k(l_i - l_R)] \quad (2)$$

Here, the wave number $k=2\pi/\lambda$, and $S_0(k)$ represents the power spectral density of the broadband light source. In the right side of formula (2), each term holds specific physical significance. The first term corresponds to the power spectral density of the reference light. The second term represents the power spectral density of each reflection point within the sample. The third term embodies the cross-power spectral density between the reflection points in the sample. Meanwhile, the fourth term stands for the cross-power spectral density between the reference mirror and each reflection point in the sample. According to the Wiener-Hinchin theorem, the autocorrelation function of the interference signal can be obtained by performing the inverse Fourier transform on the cross-power spectrum as shown in formula (3):

$$\Gamma(\Delta l) = R_R^2 \Gamma_0(\Delta l) + \sum_{i=1}^n R_i^2 \Gamma_0(\Delta l) + \sum_{i \neq j}^n R_i R_j \Gamma_0[\Delta l \pm (l_i - l_j)] + \sum_{i=1}^n R_R R_i \Gamma_0[\Delta l \pm (l_i - l_R)] \quad (3)$$

At the right end of formula (3), it is worth noting that term 1 and term 2 respectively represent the self-coherence functions of the reference light and the sample light. These components fall under the category of DC noise. Term 3, on the other hand, constitutes the mutual coherence function between the sample light of each reflection point. This term pertains to coherent noise as well as mirror noise. Meanwhile, term 4 is identified as the mutual coherence function between the reference light and the sample light of each reflection point. This particular term corresponds to the information regarding the n reflection points within the sample, along with mirror noise.

2.3. Performance metrics of OCT system

The foregoing primarily provides a brief overview of the acquisition principles underlying diverse OCT techniques. In this section, we will conduct an in-depth analysis of the key indicators pertinent to OCT and compare the performance of different OCT modalities. These indicators serve as the bedrock for the mature application of OCT systems and are of paramount importance for the subsequent *in-situ* surveillance and quality management of the laser welding process.

2.3.1. Spatial resolution

The spatial resolution of OCT can be bifurcated into longitudinal and transverse depth resolutions. The longitudinal depth resolution hinges on the coherence length of the broadband light source, with its value equaling half of that coherence length. Taking Gaussian light sources as an exemplar, if we assume the central wavelength to be λ_0 , the full width at half maximum (FWHM) to be $\Delta\lambda$, the refractive index to be n , and the coherence length to be l_c , then the longitudinal depth resolution δ_l can be computed as follows:

$$\delta_l = \frac{l_c}{2} = \frac{(2 \ln 2) \lambda_0^2}{\pi n \Delta \lambda} \quad (4)$$

The lateral resolution is determined by the size of converging spots on the sample and can be approximated by the diffraction limit formula. Suppose the focal length of the front objective lens is f , the light spot diameter on the objective lens is d , and the refractive index is n , thus the lateral resolution δ_t is as follows:

$$\delta_t = \frac{1.22 f \lambda_0}{n d} \quad (5)$$

2.3.2. Imaging depth

The imaging depth of TD-OCT is mainly determined by the scanning range of the reference mirror. And the imaging depth is related to the sampling interval of interference

spectrum, which corresponding to the resolution of the spectrometer of SD-OCT and the minimum line width of the wavelength scanning light source in SS-OCT.

Suppose the refractive-index is n , the sampling width of interference spectrum is W_λ , the corresponding sampling number is N , thus the imaging depth D of FD-OCT is.

$$D = \frac{N\lambda_0^2}{2nW_\lambda} \quad (6)$$

2.3.3. Imaging velocity

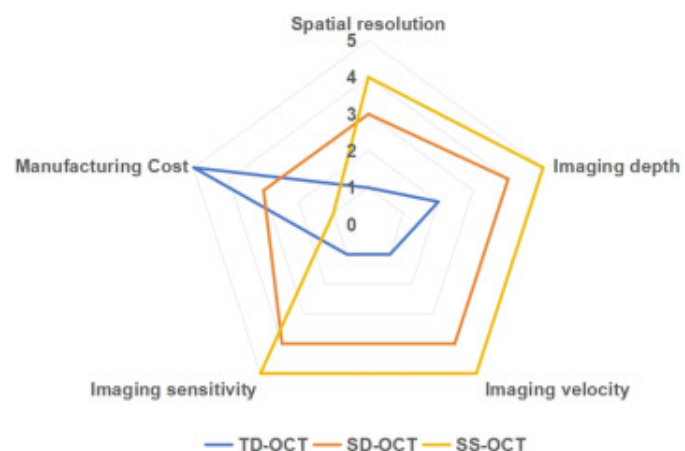
The imaging velocity of OCT is affected by the speed of depth information acquisition, transverse scanning and data processing, among which the depth information is the most important factor. Here, the depth information acquisition speed of TD-OCT is mainly affected by the depth scanning speed of the reference mirror. FD-OCT is mainly affected by the acquisition speed of interference spectrum and in which the acquisition speed of SD-OCT is mainly restricted by the readout speed of CCD, and the acquisition speed of SS-OCT is mainly restricted by the scanning speed of wavelength scanning light source.

2.3.4. Imaging sensitivity

The sensitivity of OCT can be defined as the ratio of the reflectance of the reference mirror to the minimum reflectance of the sample, which is determined by the incident light power, the reflectance of the sample, and the measured depth. And depth information acquisition speed and other decisions. In weakly scattering media, the sensitivity determines the minimum detectable light signal. In strongly scattering media, sensitivity determines the maximum imaging depth. The sensitivity of OCT can be improved by extending the signal acquisition time and reducing the depth information acquisition speed.

2.4. Comparison between the different OCT techniques

Fig. 5 presents a comprehensive performance comparison among diverse OCT techniques. In terms of spatial resolution, imaging depth, velocity, and sensitivity, the observed values are approximately in the vicinity of 5. Such proximity to this value indicates that the corresponding performance metrics are relatively favorable; conversely, values deviating further from 5 would imply less satisfactory performance. It is worth noting that the actual manufacturing cost of the OCT system is comparatively high. In the context of the evaluation, this translates to the manufacturing cost metric being close to 1, signifying a relatively significant cost burden.



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Fig. 5. Performance comparison between different OCT techniques.

According to the physical fundamentals of different OCT techniques and performance comparison results, we can found that:

- (1) TD-OCT generally apply a broadband light source, point detector and mechanical scanning mirror. The maximum sampling frequency is several hundred Hertz, the imaging speed is greatly affected by the mechanical scanning structure and the imaging speed is slow.

- (2) Compared with TD-OCT technology, SD-OCT technology greatly improves SNR and imaging speed by simultaneously detecting multiple wavelengths and resolving different depths. It has the advantages of single point detection of TD-OCT and fast imaging of SD-OCT.
- (3) SS-OCT technology applies a tunable wavelength sweep frequency laser and single-point detector, which can allow higher laser power with each discrete wavelength impinge on the sample. The SS-OCT can obtain a higher imaging sensitivity and up to MHz scanning frequency. However, the whole SS-OCT equipment including the swept laser source, photodiode balanced receiver and high-speed data acquisition card is relatively expensive compared to TD-OCT and SD-OCT.
- (4) With the above-mentioned outstanding advantages, the OCT technique can directly real-time detect the depth information during laser welding process. In fact, applying OCT technology to laser and multi-energy field composite processing technologies such as high-power (6–10kW) laser-arc hybrid welding has broader application prospects. Although SS-OCT is relatively expensive and has not yet achieved large-scale application, it offers higher imaging speed and greater penetration depth (10–15mm) compared to SD-OCT (less than 10mm). With the reduction in the cost of key optical components, SS-OCT is the development trend in the future.

Next section, we will analyze and discuss the main application and potentials of OCT technique in laser welding process.

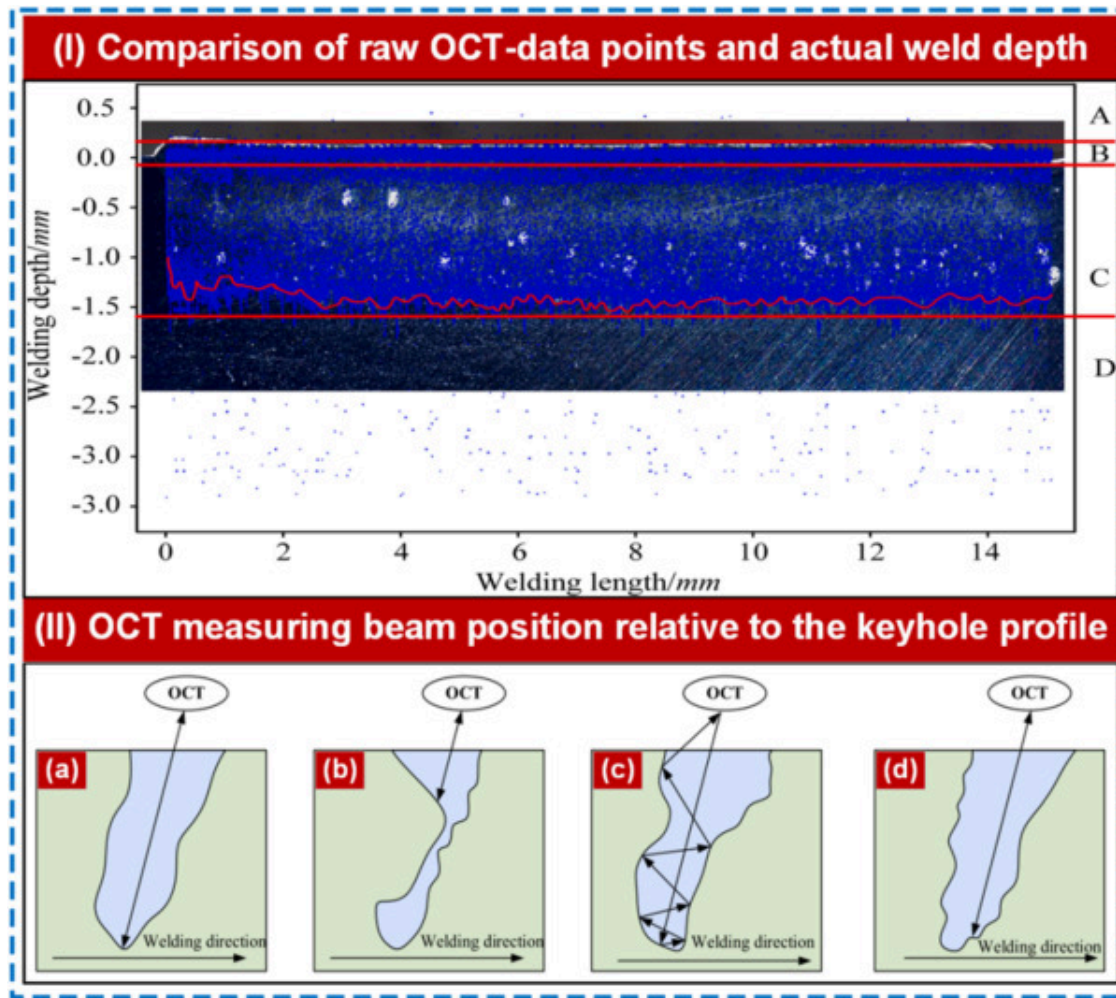
3. OCT-data processing and precision analysis

3.1. Motivation of OCT-data processing

Within the intricate context of the LKW process, the attainment of a precise and dependable OCT measurement of the vapor-filled capillary depth, assumes a position of paramount importance for the prediction and assurance of welding quality. However, this task has progressively metamorphosed into an increasingly formidable challenge, primarily due to the harsh operating environment that is rife with weld fume, metal spatters, as well as intense light and thermal radiation. Concurrently, OCT signals are profoundly influenced by the complex and dynamic behaviors of the melt pool and keyhole. This influence introduces an inherent complexity in the processing and interpretation of the raw OCT data, as the signals are subject to rapid and unpredictable fluctuations. Consequently, the OCT signal frequently exhibits a relatively low signal-to-noise ratio (SNR), which necessitate the development of advanced signal processing techniques and innovative measurement strategies to enhance the accuracy and reliability of OCT measurements in the demanding environment of laser welding.

As a recently research indicates [40], a distinct correlation can be discerned between the weld longitudinal section and the acquired OCT-data (depicted by blue points). As illustrated in Fig. 6(I), the bottom of the densely scattered area exhibits a remarkable congruence with the actual penetration depth. Subsequently, the OCT-data can be segmented into four domains, namely A, B, C, and D. The points falling within domains A and D predominantly constitute noise regions and thus necessitate screening out prior to the extraction of the capillary depth. The points within domain B chiefly embody the surface depth information, which is highly susceptible to perturbations caused by alterations in surface roughness and subsequent reflections of the OCT beam. Likewise, the points within domain C pertain to the actual keyhole depth information subsequent to the entry of the OCT probe beam into the keyhole. As shown in Fig. 6(II)-(a), the probe beam penetrates to the bottom of the keyhole and reflects back. Additionally, the scattering information beneath the actual depth profile can be ascribed to the probe light reaching the keyhole bottom yet being refracted at the keyhole wall, thereby engendering a longer light path with multiple reflections, as portrayed in Fig. 6(II)-(c). The principal factors responsible for the scatter points above the depth profile are laser incidence

deviation, keyhole collapse, or keyhole-induced pores, as exemplified in Fig. 6(II)-(b) and (d).



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Fig. 6. The distribution characteristics of the acquired OCT data under various keyhole dynamics: (I) comparison of acquired OCT-data and weld depth; (II) OCT beam position relative to the keyhole profile [40].

Overall, the aforementioned phenomena can give rise to a more extensive distribution of complex and discrete points, further complicating the precise extraction of the actual melt depth. This, in turn, will impede its application in real-time monitoring and quality control. Moreover, the performance of the OCT system is liable to be circumscribed by imaging resolution and depth perception, especially when confronted with highly reflective materials in laser welding applications. Hence, it becomes imperative to accurately process the OCT-data to extract valuable melt-depth information.

3.2. OCT-data processing techniques

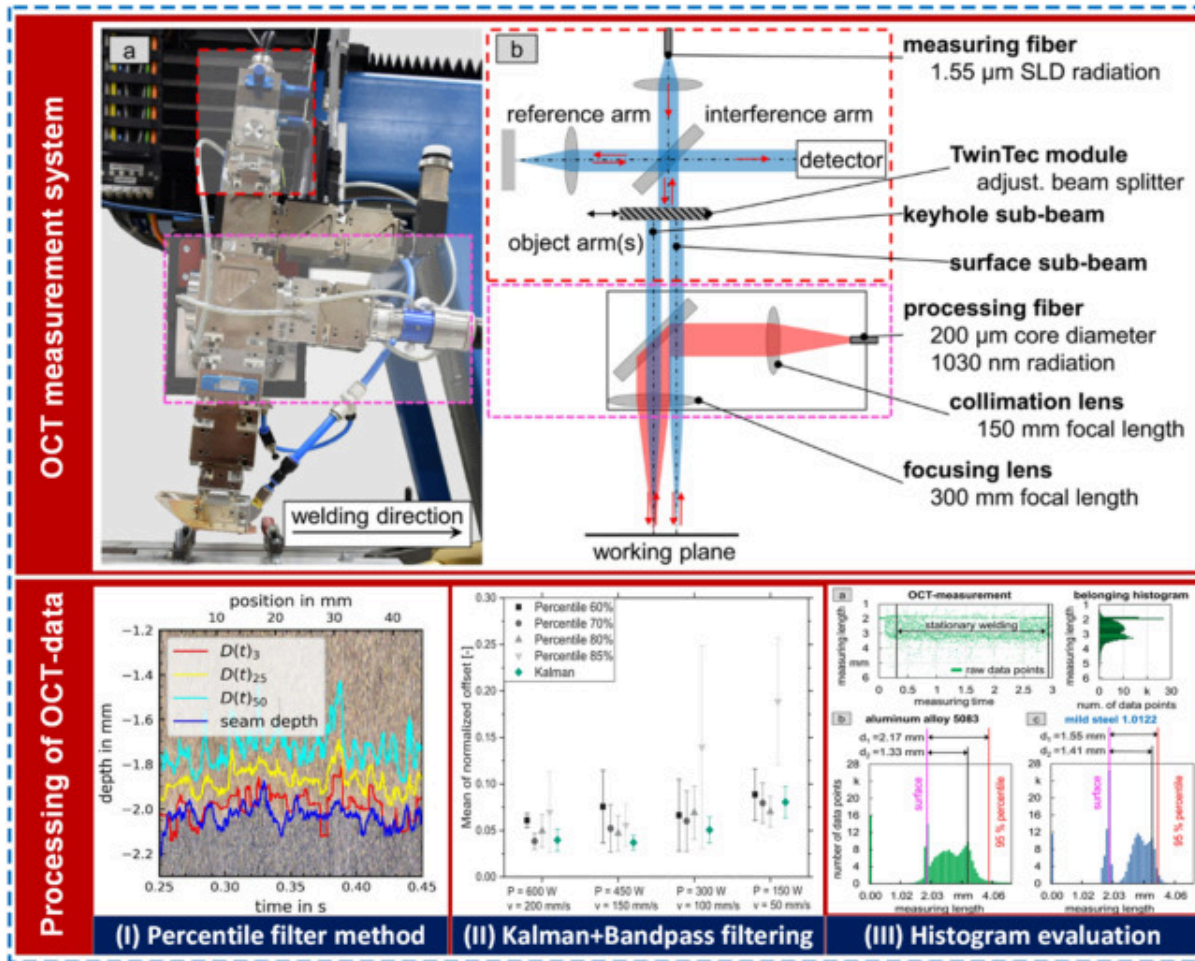
In order to tackle the prominent challenges concerning noises and the extraction of melt-depth within raw OCT data, contemporary research predominantly centers around the development of cutting-edge OCT data processing algorithms, as outlined in [Table 3](#). In the latest developments, heuristic methods have been widely adopted in these filtering processes to extract the actual penetration depth from the raw signals. Boley et al. [41] initially put forward a two-step statistical assessment approach integrating Poisson distribution and percentile filtering. As depicted in [Fig. 7\(I\)](#), based on the OCT measurement system, it was ascertained that the Poisson distribution technique is capable of efficiently segregating the noises from the initially acquired data by scrutinizing the frequency distribution. Subsequently, a running percentile filter, with a window length spanning from 5 to 20ms and a percentile ranging from 2% to 5%, was implemented to extract the depth information, aligning it with the actual weld seam depth. Moreover, Hollatz et al. [42] adopted a combination of Kalman and band-pass filters, which significantly enhanced the OCT measurement data and enabled a more accurate representation of the weld depth. To further augment the efficiency and precision of OCT-data processing methodologies, certain domestic scholars [40,43,44] have proposed a hybrid approach. This entails the utilization of Density-Based Spatial Clustering of Applications with Noise (DBSCAN) or Hierarchical-DBSCAN algorithms to filter the raw OCT signals, complemented by suitable percentile filtering for extracting the melt depth curve. These approaches furnish more accurate parameters for the *in-situ* prediction of weld defects and quality control. (See [Table 3](#).)

Table 3. Summary of different OCT data processing methods (Thickness d , Sample rate f_s , Central wavelength λ_0 , Probe beam diameter d_{pb}).

Processing methods	OCT configuration	Materials	Processing results	Ref.
- Normalized Poisson distribution - Percentile filter	$f_s=70\text{kHz}$ $d_{pb}=70\mu\text{m}$ $\lambda_0=1540\text{nm}\pm 20\text{nm}$	Mild steel Aluminum alloy	Average error <5%	[41]
- Histogram analysis/evaluation - Frequency distribution	$f_s=70\text{kHz}$ $d_{pb}=70\mu\text{m}$ $\lambda_0=1540\text{nm}\pm 20\text{nm}$	Aluminum alloy 5083 Mild steel 1.0122 $d=6\text{mm}$	Absolute deviation is between 0.1 and 0.27 mm, Resulting percent error $\leq 10\%$.	[45]
- Kalman filter - Bandpass filter	$f_s=70\text{kHz}$ $d_{pb}=80\mu\text{m}$ $\lambda_0=1540\text{nm}\pm 20\text{nm}$	Stainless steel 1.4301 $d=1.5\text{mm}$	The error of data points calculated using Kalman filter and bandpass filter falls within the lowest range.	[42]

Processing methods	OCT configuration	Materials	Processing results	Ref.
- Intensity accumulation within ROI - Scanning OCT measurement line	$f_s=70\text{kHz}$ $\lambda_0=840\text{nm}\pm 20\text{nm}$	Pure copper $d=5\text{mm}$	The utilization of intensity accumulation leads to the attainment of the lowest root mean square error (RMSE).	[46]
- Kernel density estimation (KDE) - Moving window	$f_s=70\text{kHz}$ $d_{pb}=80\mu\text{m}$ $\lambda_0=1540\text{nm}\pm 20\text{nm}$	Aluminum 1050 $d_1=450\mu\text{m}$ Nickel-plated copper $d_1=350\mu\text{m}$	Increase OCT measurement accuracy of about 50%.	[48]–[49]
- Density-Based Spatial Clustering of Application Noise (DBSCAN) - Percentile filter	$f_s=80\text{kHz}$ $\lambda_0=840\text{nm}\pm 45\text{nm}$	Q235 mild steels, Mild steel Q235 Al 6061, Cu T2	Average error < 5%	[43]–[44]
Hierarchical Density-Based Spatial Clustering (HDBSCAN)	$f_s=80\text{kHz}$ $\lambda_0=1540\text{nm}\pm 20\text{nm}$	304 stainless steel	Average error < 5%	[40]

Processing methods	OCT configuration	Materials	Processing results	Ref.
- Percentile filtering - Exponential moving average method				
- Standard deviation (STD) features (Ensemble) Empirical mode decomposition	$f_s=250\text{kHz}$ $\lambda_0=840\text{nm}$	Aluminum 6061 and 7075 316 stainless steel	It can quantify the degree of keyhole fluctuation more accurately and reliably.	[50] – [51]



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Fig. 7. Different data processing techniques based on OCT sensor: (I) Percentile filter method [41]; (II) Kalman + bandpass filtering [42]; (c) Histogram evaluation [45].

Table 3. Summary of combination of OCT with various ML techniques.

Scenarios	Machine learning	Detection object	Ref.
Laser beam welding of Copper Hairpin Joints	SVM, XGBoost, LightGBM, MLP, CNN, MLP	Hairpin welding defect detection	[106]
Laser beam welding of High-strength aluminum alloys AA6082	Beta-Variational-Autoencoder (β -VAE)	Prediction of the weld depth	[108]
Laser lap welding of thin stainless steel plates	Long short-term memory (LSTM) network		[51]
Laser beam welding of Copper alloy C10200	Artificial neural networks (ANN)	Prediction the weld seam quality (good/poor)	[63]
Spot-in-Spot laser beam welding of Pure nickel (Ni 2.4068)	Artificial neural networks (ANN)	Identifying direction of weld path deviations	[107]
Laser welding of aluminum alloys	Deep belief networks (DBN) Sparse auto-encoder (SAE) Artificial neural networks(ANN)	Online porosity prediction	[67]- [68,110]
Aluminum-copper dissimilar metal ARM laser welding	Back-propagation neural network (BPNN)	Prediction of seam width at the interface (SWI)	[109]
Q235 mild steel Laser lap welding	Hidden Markov Model (HMM)	Quantitative evaluation of keyhole stability (open/collapse)	[111]

Scenarios	Machine learning	Detection object	Ref.
	Convolutional Neural Network (CNN)	Prediction of seam width at the interface (SWI) and weld depth	[112]

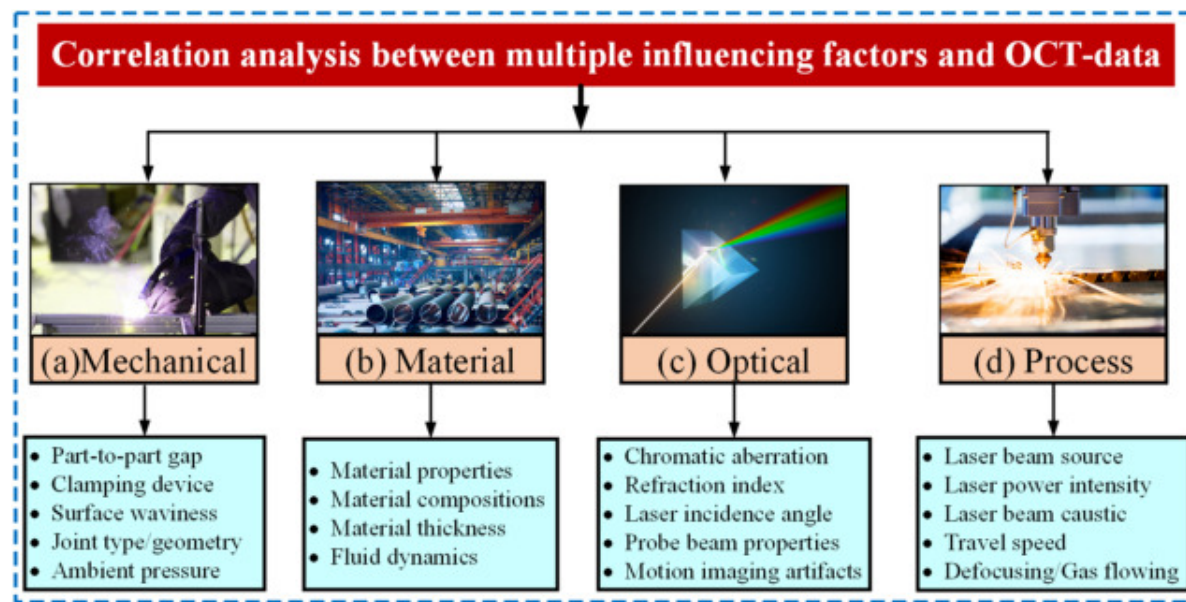
Notwithstanding the fact that the accuracy of OCT measurement has been substantially enhanced through appropriately parameterized percentile filtering techniques, there remains both the necessity and significance of calibrating/adjusting the OCT filter parameters using copious experimental data. Additionally, issues such as reproducibility and variations in the applicable percentile values are complicating its practical application. This is due to their dependence on process parameters and the challenges encountered in noise removal procedures, which add to the overall complexity. Consequently, Mittelstädt and his colleagues [45] put forward a new depth-assessment methodology. This approach makes use of histogram evaluation that hinges on the frequency distribution of OCT-data. Notably, the OCT-data exhibits a particular pattern, featuring a local maximum value that correlates with the keyhole depth, as illustrated in Fig. 7(III). In parallel, Will et al. [46] suggested an efficient intensity accumulation technique for gauging weld depth during copper laser welding, with OCT measurement beams being scanned laterally. Additionally, Pordzik et al. [47] deduced specific traits from the histographic analysis, which were employed to delineate the ambiguity in depth measurement and the variation in the welding process.

In order to gain a more profound understanding and disclose the intricate dynamics of the keyhole, it is imperative to adopt alternative and advanced data processing methodologies that can offer a more conclusive approach. Accordingly, Sokolov and his collaborators [48,49] devised a kernel-density-estimation (KDE) technique employing a Gaussian kernel for the OCT time-series signals. By doing so, they estimated the nonparametric statistical density function, which is capable of characterizing the shape and depth of the keyhole. To precisely gauge the extent of keyhole fluctuation, Ma and

Shu et al. [50,51] utilized a sliding window to scan the raw OCT signals and compute the standard deviation (STD) features. Subsequently, they employed the empirical mode decomposition (EMD) to eliminate noise interference and high-frequency constituents, retaining only the low-frequency approximate trends, thereby obtaining the local characteristics of the OCT signals.

3.3. Correlation analysis between influencing factors and OCT-data

The OCT system enables a direct inline measurement of the vapor capillary depth in the course of the dynamic laser welding process. In order to achieve a reliable interpretation of the OCT measurement results, a substantial number of influencing factors need to be taken into account. As depicted in Fig. 8, the existing multiple influencing factors are classified into four categories: mechanical, material, optical, and process. These factors can directly influence the accuracy and dependability of OCT sensing. Therefore, it is imperative to analyze the correlation between the various influencing factors and the OCT data.



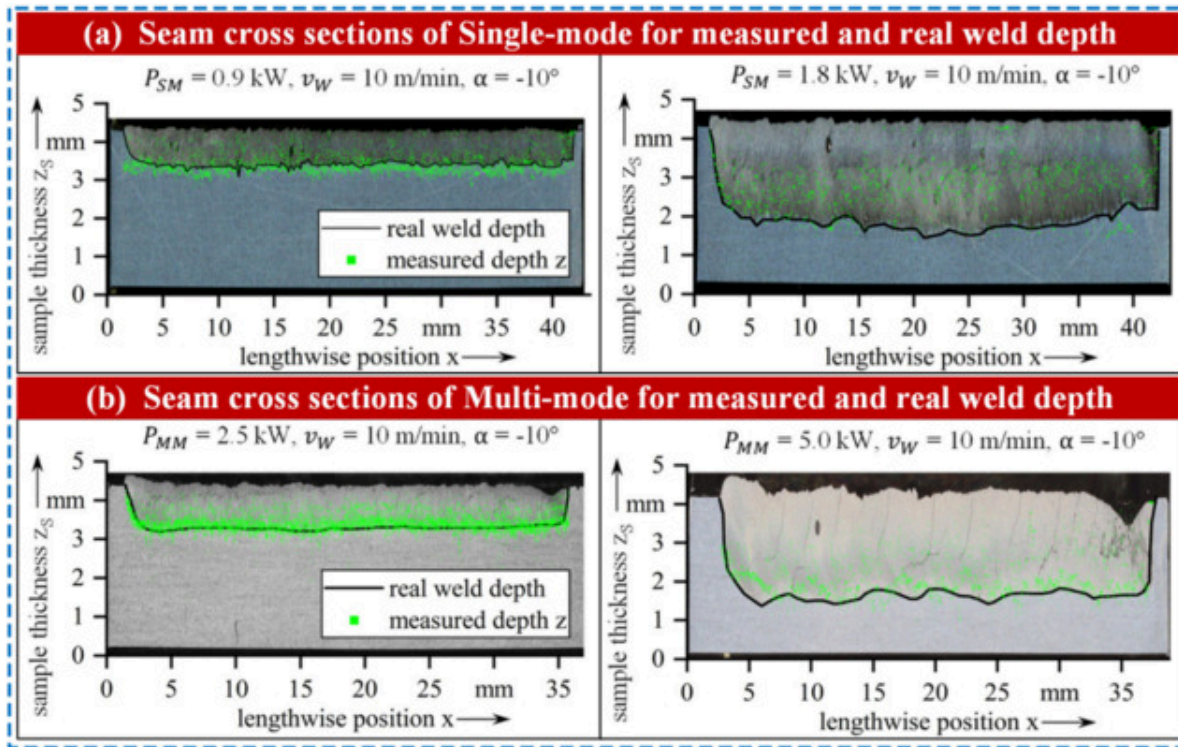
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Fig. 8. The correlation analysis between multiple influencing factors (mechanical, material, optical and process) and OCT-based depth data.

Particularly, the mechanical factor arises from random fluctuations related to clamping or fastening fixtures, part-to-part clearances, joint configurations, and ambient pressure. The material factor pertains to differences in material thickness, composition, and characteristics. It should be emphasized that the unique characteristics exhibited by OCT measuring signals when dealing with different materials can be attributed to the material-specific geometry of the keyhole. In terms of the optical factors, these encompass alterations in the refractive index within the plasma plume that hovers above the molten pool, possible chromatic aberrations that arise from the concurrent passage of the OCT measuring beam and the laser beam through the same lens during their interaction, and also motion imaging artifacts. The process factor encompasses the laser beam source, power intensity, travel speed, and weld depth. Research reported in [49] concluded that the combined effect of these influencing factors results in the fact that the position of the OCT measuring beam is not standardized and must be adjusted for each specific welding setup.

Schmoeller et al. [52] conducted a comprehensive investigation into the influences exerted by process parameters, workpiece materials (such as Al and Copper), the incidence angle of the probe beam (ranging from -20° to $+20^\circ$), and the weld joint geometry on the OCT measuring signals. Through this analysis, the OCT detection accuracy could be evaluated, thereby facilitating a better validation of the precise measurement capabilities and physical properties of the OCT technique. Fig. 9 presents a comparison of the different OCT measurement results in the context of process variations, which include the utilization of different laser beam sources (single-mode and multi-mode) as well as variations in laser power. It was discovered that the application of the OCT technique enables *in-situ* and real-time monitoring of the weld depth, regardless of whether a multi-mode or a highly focused single-mode laser beam is employed.



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Fig. 9. Longitudinal cross sections of metallographic samples obtained from single-mode and multimode welding for comparing the values measured by OCT with the actual weld depth [52].

Through iterative adaptation of the keyhole geometry in simulations based on the genetic algorithm (GA), Schmoeller et al. [53] were also able to ascertain and quantify the dependence of the liquid melt-pool thickness beneath the keyhole on the process parameters. This approach effectively addresses the calculation of the deviation (a geometric quantity) between the measured keyhole depth and the seam depth. Furthermore, the deviation value serves as a crucial parameter for interpreting keyhole depth measurements in relation to the seam depth. By implementing a commercially

available ICI (Inline Coherent Imaging) system predicated on OCT, Patterson et al. [54] delved into the internal relationship linking laser weld parameters and the vapor capillary (keyhole). The measurement results unequivocally demonstrated that laser parameters exert a significant impact on keyhole morphology and the position of the keyhole root relative to the laser beam.

In addition, Möbus et al. [55] devised a laser beam welding setup enabling the coaxial combination of two distinct wavelengths of laser sources. Subsequently, they examined the influence of specific intensity distributions on the accuracy of OCT measurements. Mattulat and his colleagues [56,57] carried out bead-on-plate welding experiments under diverse ambient pressure conditions and using different materials. The aim was to evaluate the opening of the keyhole and the degree to which the OCT probe beam could access it. The findings revealed that the improved measurement clarity at lower ambient pressures was not due to increased access of the OCT probe beam to the keyhole. Instead, it was a result of the changed conditions inside the keyhole. In comparison to steel plates, experiments with aluminum plates provided further evidence of the positive impact of reduced ambient pressure on the ability to detect the bottom of the keyhole. Regarding mechanical variations, a study [58] investigated the welding of hidden T-joints. In these experiments, the laser path had angular and parallel offsets relative to the center line of the web sheet. At the same time, OCT signals were recorded. These signals were subsequently analyzed to determine whether OCT monitoring was suitable for predicting an imminent misalignment between the laser beam and the web sheet before it actually happened.

To date, certain research has established the internal connections between the multiple influencing factors (mechanical, material, optical, and process) and OCT data, which can offer valuable guidance for OCT-data processing and comprehension. However, the laser welding process inherently possesses characteristics of complexity and dynamic alteration, necessitating the consideration of all potential factors, especially the interaction between the OCT probe beam and the behaviors of the melt pool/keyhole. In fact, a more profound understanding of the keyhole “forming-growing-oscillation”

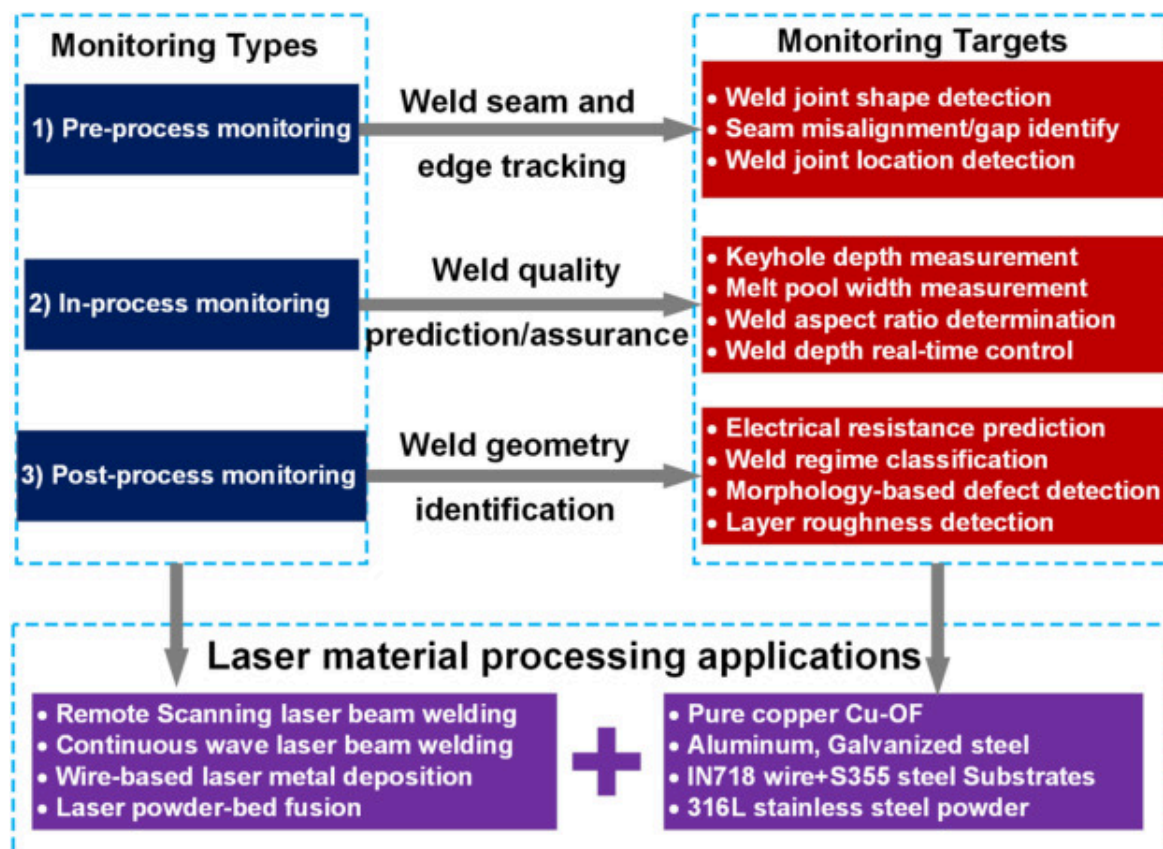
mechanisms, contingent upon the interplay between the laser source and materials, is of paramount importance. This deeper insight will enable better analysis and extraction of the key characteristics of OCT signals.

4. All-process OCT monitoring and quality assurance

After conducting OCT-data processing and precision analysis, OCT unlocks new possibilities for modern laser manufacturing systems, enabling comprehensive all-process monitoring. It offers a streamlined, trendsetting quality assurance approach that caters to cost-efficient, high-performing, and high-throughput laser welding applications. In this part, we will explore three principal forms of OCT monitoring:

- 1) **Pre-process monitoring:** Specializing in tracing the joint gap and workpiece geometry and spotting any misalignments. This is crucial for enhancing weld quality as it allows for preemptive adjustments.
- 2) **In-process monitoring:** Capturing the keyhole depth, which provides real-time insights that are invaluable for predicting weld penetration and overall quality, enabling immediate corrective actions if needed.
- 3) **Post-process monitoring:** Retrieving the shape and geometry of the final weld, which is useful for quality assessment and process validation, ensuring that the end product meets the required standards.

These different OCT monitoring phases can be carried out concurrently, depending on the specific requirements of the laser processing operation. [Fig. 10](#) succinctly presents the diverse monitoring types along with their corresponding objectives in the context of laser processing applications.



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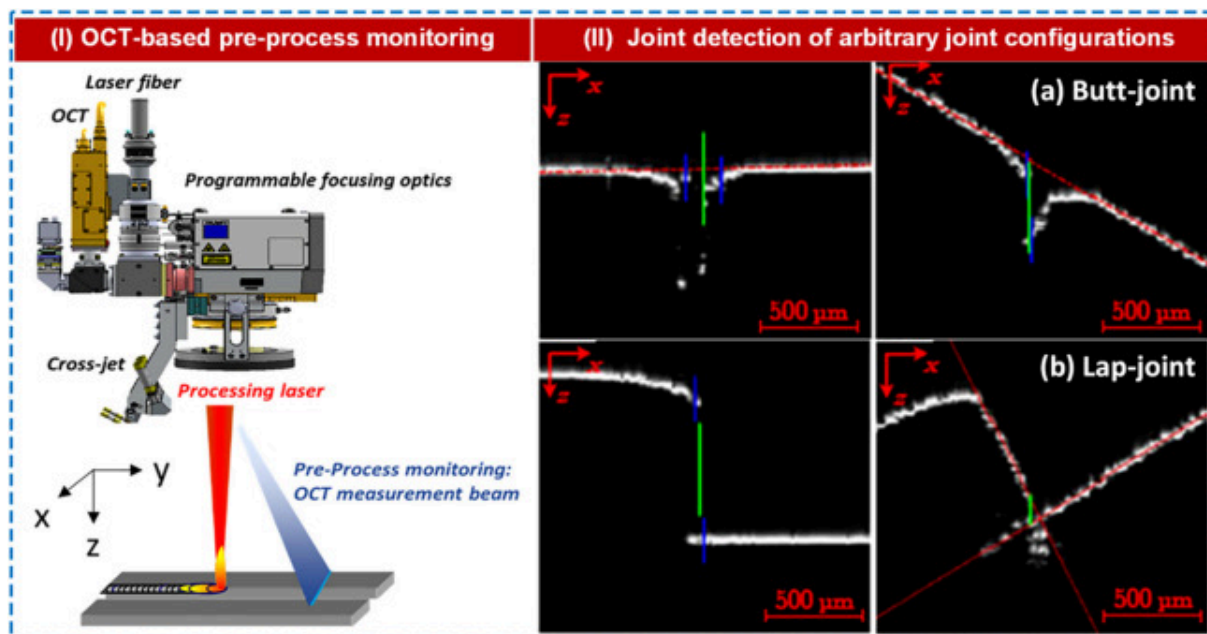
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Fig. 10. Various monitoring types and the corresponding targets in laser processing applications.

4.1. Pre-process OCT monitoring

Generally speaking, Pre-process monitoring makes it possible to measure the joint position, which is essential for accurately positioning the processing laser and verifying the geometry of the workpiece. Cutting-edge seam tracking techniques predominantly hinge on monochromatic cameras or laser triangulation. Typically, these methodologies make use of line segmentations to pinpoint the joint position. The OCT interferometric

measurement approach has, however, unearthed novel image characteristics during the seam tracking procedure. Will et al. [59] incorporated an extra scanner to achieve highly dynamic positioning of the OCT probing beam, decoupling it from the laser beam. In doing so, they were able to isolate specific noise components and identify features pertinent to image processing in the context of seam tracking applications. Subsequently, they proposed two forms of suitable image-processing strategies for spotting joints of various configurations, *e.g.* butt-joints or lap-joints, as depicted in Fig. 11. Furthermore, Stadter et al. [60] adopted a similar 3D scanner mechanism to acquire the surface topography information of the laser-exposed area and its surroundings. They then verified the OCT measuring solution for edge tracking in remote laser welding (RLW) when dealing with materials like aluminum, copper, and galvanized steel. Dupriez et al. [29,30] developed a commercial OCT system in collaboration with Lessmüller Lasertechnik GmbH. This system aims to facilitate the reliable and stable production of nearly all types of laser-welded joints (*e.g.* flange joint, fillet joint, radial butt joint, and V-joint). It can effectively control the welding position, incidence angle, and gap size, ensuring that gaps as large as 0.8mm can be successfully bridged without the need for complicated laser beam oscillation methods.



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Fig. 11. Characteristics for the image processing of OCT images in seam tracking for laser welding: (I) The experimental configuration encompasses a programmable scanning focusing optic accompanied by a cross-jet, a processing laser with a laser fiber, and an OCT apparatus. (II) Illustrative image processing procedures for identifying the joint position are shown in [59].

The aforementioned studies demonstrate that, in contrast to traditional vision and laser triangulation techniques, OCT-based Pre-process monitoring boasts a highly prominent advantage in terms of omnidirectional high-resolution online seam tracking.

Furthermore, it has the capacity to provide three-dimensional quantitative details about the surface topography in front of the laser processing zone. As a result, OCT emerges as an unparalleled technique for laser welding applications. It can make substantial progress in facilitating direct, accurate, and reliable product line traceability, thus guaranteeing

that weld quality standards are fulfilled. Looking to the future, forthcoming research will concentrate on applying an evolutionary algorithm to optimize image processing. This methodology is anticipated to more effectively enable real-time seam tracking during the Pre-processing monitoring stage.

4.2. In-process OCT monitoring

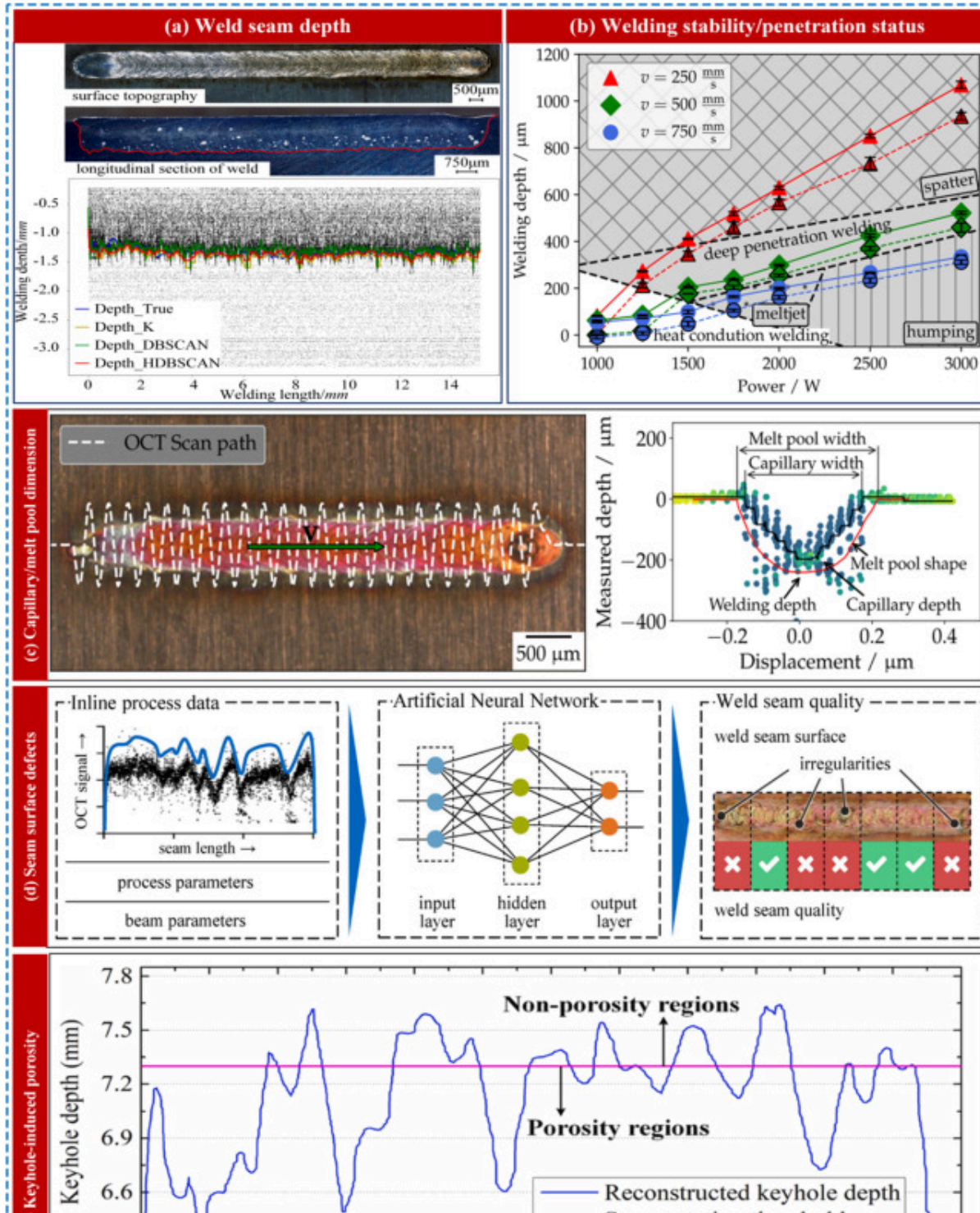
Subsequent to the completion of Pre-process OCT monitoring, probing the interior of the keyhole during the laser welding process holds significant importance as it enables the evaluation of process stability and the determination of the final weld quality. In Section 4.2, we will explore the OCT's capacity for predicting and ensuring weld quality.

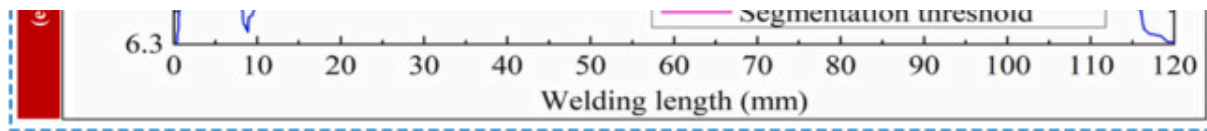
Subsequently, we will analyze and derive both the potential advantages and the existing limitations associated with in-process monitoring.

4.2.1. Quality prediction

Generally speaking, the main aspects of laser welding quality encompass the following:

(a) weld seam depth; (b) welding stability and penetration status; (c) capillary/melt pool dimensions; (d) seam surface defects; (e) keyhole-induced porosity. The application of OCT monitoring methods in quality prediction can be outlined in [Fig. 12\(a\)-\(e\)](#).





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Fig. 12. The application of OCT monitoring methods in quality prediction: (a) weld seam depth [40]; (b) welding stability and penetration status [61]; (c) moltpool dimension [61]; (d) seam surface defects [63]; (e) keyhole-induced porosity [50].

When integrated with a suitable OCT-data processing methodology, OCT mainly concentrates on gauging the weld depth by situating a solitary measurement beam within the keyhole. A laterally scanned measurement beam presents the opportunity to measure both the width of the keyhole and that of the melt pool. OCT emerges as a promising technology, potentially allowing for the concurrent measurement of the weld penetration depth and the geometry of the melt pool through the perpendicular scanning of the measurement beam. Subsequently, Beck et al. [61] validated the applicability and dependability of a scanning OCT system for quantifying the capillary depth during copper micro welding with a green laser. It becomes evident that measuring the capillary/keyhole depth can directly ascertain the aspect ratio and evaluate the welding quality status, such as heat conduction, deep penetration, spatter, and humping. By overlaying the welding motion with a spatial movement perpendicular to the welding direction, supplementary details like the widths of the capillary and the melt pool can also be procured. Moreover, Will et al. [62] utilized a scanned OCT measurement beam to monitor keyhole fluctuations and pinpointed the essential key features for forecasting the welding status, which encompasses stable deep penetration welding, unstable deep penetration welding, and heat conduction welding. They further explored discussions concerning the OCT measurement outcomes, centering on aspects like the distinguishability of the welding status, the influence of laser parameters, and the limitations of this measurement technique. After scrutinizing the results related to seam

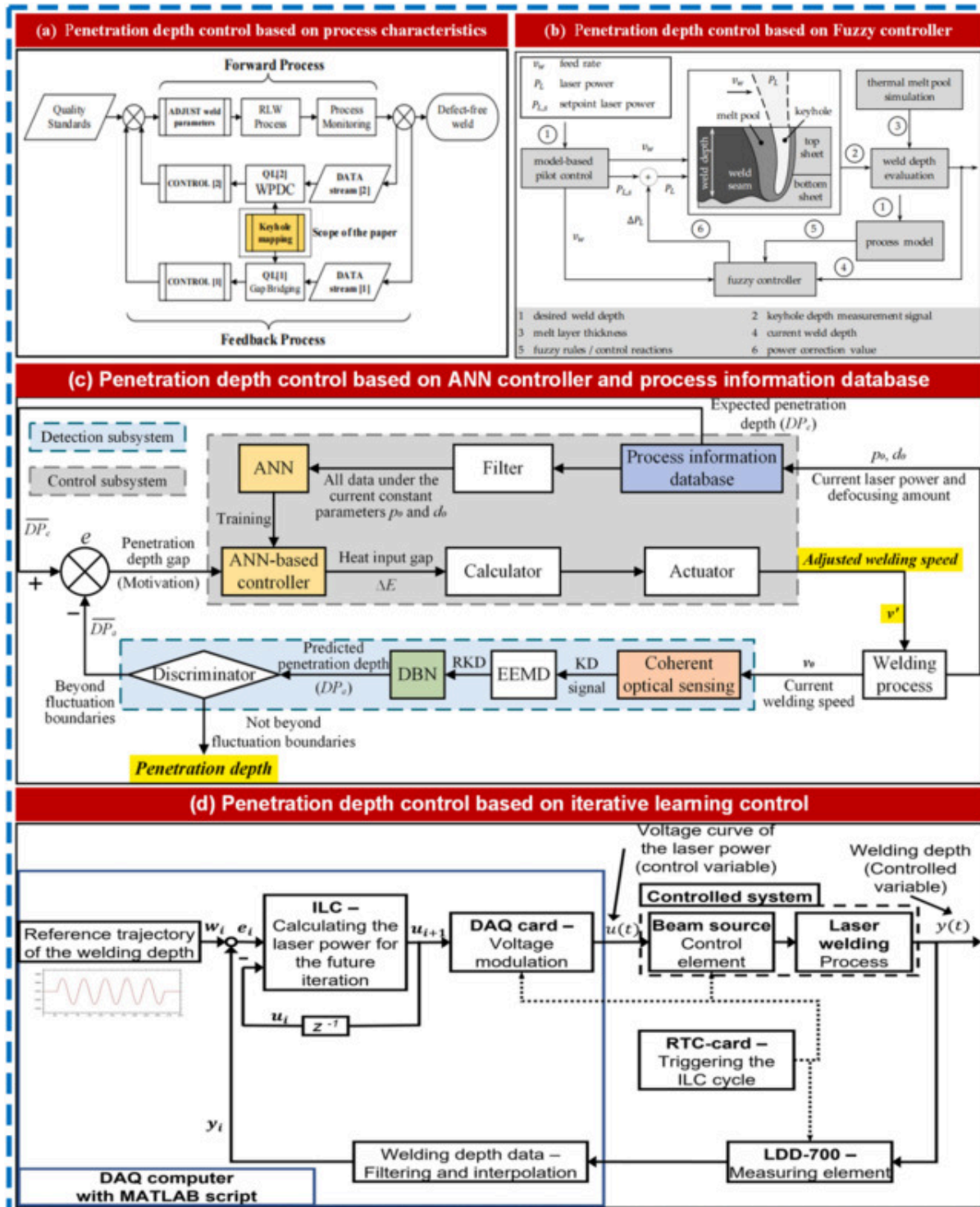
surface quality, Stadter and his team [63] adopted a machine learning approach to construct and elucidate the intrinsic correlation between the OCT-based depth signal and the surface quality. In so doing, they accentuated the importance of the critical information encapsulated within the OCT signal regarding the laser welding process phenomena that impact the surface quality.

Apart from the aforementioned predictions concerning weld surface quality (such as pool width and surface defects), other scholars have also carried out several studies on porosity detection during laser welding by utilizing OCT depth data. Webster et al. [64] verified that the OCT system can detect the formation of pores by varying the power and transitioning between two significantly different penetration depths. Boley et al. [65,66] also argued that the specific areas in OCT signals with higher noise levels might correspond to the locations of porosity. However, the OCT system might not be sufficiently sensitive to precisely distinguish the possible locations of pore formation. Instead, such locations may only be indirectly monitored or inferred based on the detection and identification of keyhole instability. To further confirm the above viewpoint, Ma et al. [67] from Huazhong University of Science and Technology recently proposed an online identification method for the regions of keyhole-induced porosity based on the OCT method. They carried out ensemble empirical mode decomposition (EEMD) on the keyhole depth (KD) signal that was measured by the OCT sensor. After that, by setting the segmentation threshold for the reconstructed keyhole depth (RKD) signal, the regions with porosity and those without porosity could be efficiently distinguished.

4.2.2. Quality assurance

Laser welding quality assurance plays a vital role in achieving excellent and consistent weld formation. The fundamental prerequisites for a high-quality weld seam render the application of a closed-loop control system indispensable in fully-automated precision machining applications, such as those in battery cells and car-body components. Specifically, when a novel and direct OCT measurement method is employed to obtain

direct geometric quantities, for instance, the weld penetration depth, a substantial control potential can be tapped. This can be achieved by automatically adjusting the laser power or other parameters, based on the calculated deviation between the target curve and the measured value of the penetration depth. In this respect, some scholars have conducted research on quality control methods that rely on the OCT monitoring method. Through adjusting different control strategies from two aspects, namely (a) process characteristics and (b) machine learning and intelligent control, these methods are able to assist in attaining the optimal weld penetration depth that falls within the necessary limits, which as illustrated in [Fig. 13](#).



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Fig. 13. Different in-process Weld Penetration Depth Control (WPDC) algorithms: (a) process properties-based controller [49]; (b) fuzzy logic-based controller [69]; (c) ANN-based controller [133]; (d) iterative learning-based controller [70].

In the earlier years, Kawahito et al. [68] proposed the notion that by having a precise prior understanding of how process parameters impact welding quality, the effectiveness and accuracy of penetration depth control could be significantly enhanced through modifying the control framework. Subsequently, Sokolov M et al. [49] devised an OCT-based Weld-Penetration-Depth-Control (WPDC) strategy. In this approach, they capitalized on process characteristics as a crucial element to actualize adaptive penetration depth control during Remote Laser Welding (RLW) of aluminum in the form of fillet lap-joints, taking into account the variation in part-to-part gaps. The newly proposed penetration depth control method, which hinges on process characteristics, showcases the outcomes of keyhole mapping under diverse part-to-part gap scenarios. These results provide significant revelations for forthcoming research endeavors that strive to attain fully automatic closed-loop weld penetration depth control.

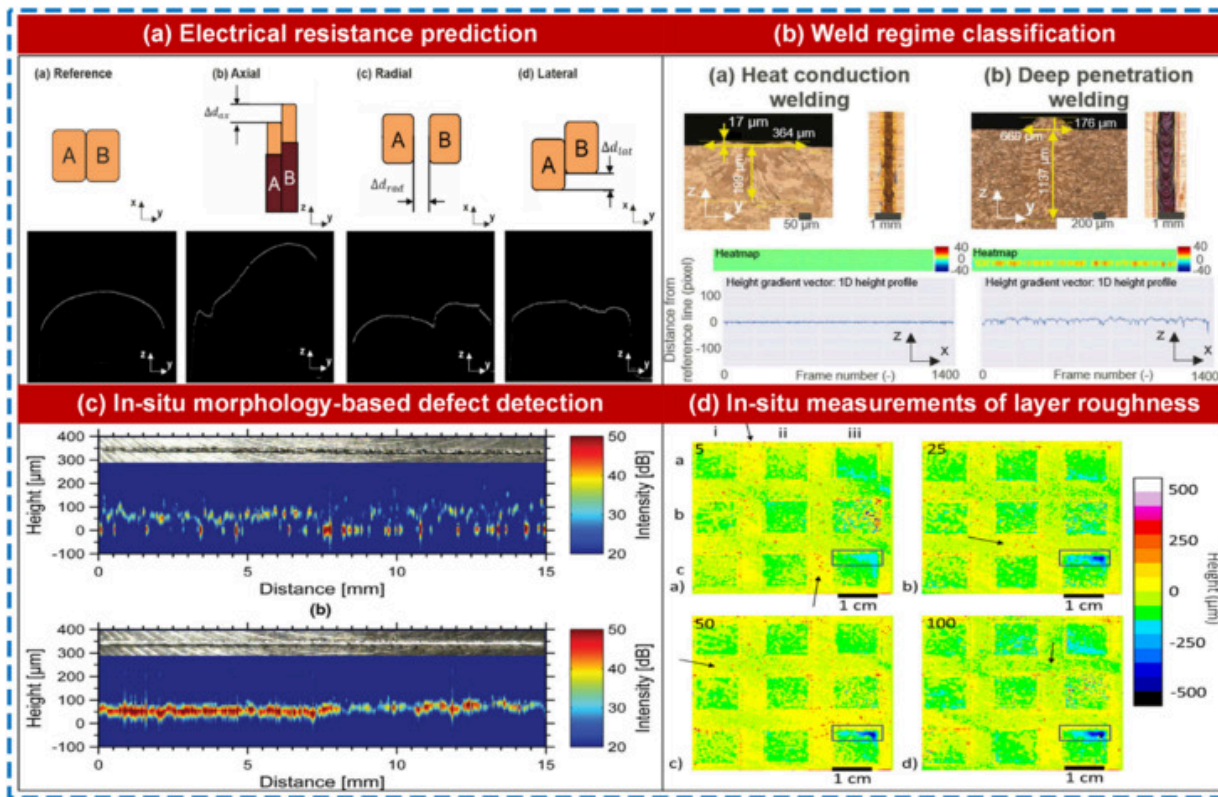
Furthermore, Maximilian et al. [69] systematically explored the physical differences between the keyhole and the weld depth, and then applied an inline weld depth fuzzy intelligent control system, which was developed based on the knowledge obtained from machine learning techniques. After establishing the process information database, Ma and his collaborators [133] put forward an ANN (artificial neural network)-based control strategy aimed at regulating the penetration depth by adjusting the welding speed. Later on, the control performance was examined and validated through the detection of penetration depth with the help of the OCT system. Hence, the proposed closed-loop feedback control can provide an automated adjustable capability to ensure the attainment of a desired penetration depth. Despite the above linear weld depth control with a fixed

optics system, Spurk et al. [70] demonstrated the development and evaluation of a weld penetration depth control system. They achieved this by conducting welding with or without spatial power modulation on steel workpieces while following a sinusoidal trajectory with the galvanometer scanning optics.

However, in practical industrial applications such as those in vehicle bodies and power batteries, galvanometer scanning optics are favored for high-intensity and high-speed laser radiation with a small focal diameter. One of the keys to achieving weld depth control using a scanning optics system is to devise a calibration method for the measurement system, which enables the accurate positioning of the OCT point across the entire processing field of the scanning optics. Moreover, it is of great importance to verify the capacity of the existing control algorithms to respond to unexpected external disturbances. Additionally, developing a multi-parametric parallel or intelligent control technique should be a focus in future studies.

4.3. Post-process OCT monitoring

The surface of the weld seam plays a pivotal role in the nondestructive examination of welds to detect imperfections and defects. The topographical details of a weld seam carry valuable information related to quality characteristics like humping and spatter. To effectively utilize the surface topographical data obtained from the welding bead, data processing is essential. In the Post-process monitoring stage, the focus lies on identifying subpar weld quality by observing the geometry of the solidified weld, which as summarized in Fig. 14(a)-(d).



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Fig. 14. Post-process OCT monitoring for various laser welding applications: (a) electrical resistance prediction [71,72]; (b) weld regime classification [73,74]; (c) morphology-based defect detection [75]; (d) measurement of layer roughness [77].

Thomas et al. [71,72] applied the FRESH (feature extraction based on scalable hypothesis tests) algorithm to identify a quality-relevant feature. Their aim was to predict the electrical resistance of pure copper hairpin (Cu-ETP) welds based on surface topographical OCT data. Moreover, they further demonstrated the feasibility of the FRESH algorithm for classifying the quality of copper laser welding, such as spatter, deep penetration welding, humping, and heat conduction welding [73,74].

In the field of laser additive manufacturing, the OCT system boasts several advantages, including rapid sampling, high precision, and high reliability. It can be utilized for *in-situ* measurement of the surface height of each print layer, enabling accurate prediction of key indicators like weld geometry, roughness, and surface defects. Regarding this matter, Stehmar and his colleagues [75] created a SD-OCT system and incorporated it into a wire-based laser metal deposition (LMD-w) process by means of suitable optics. This integration allows for the accurate measurement of surface geometries with high aspect ratios, reflectivity, or roughness. More importantly, compared to traditional laser triangulation, OCT can achieve a much higher spatial resolution.

This particular trait renders an LMD-w process equipped with OCT post-process monitoring especially apt for the highly exacting production settings of small-scale products. Given that the stability of the melt pool can act as a prime indicator of the resultant track quality, Kanko et al. [76] devised a high-speed imaging rate and time-resolved OCT measurement methodology for observing the morphology of the melt pool during the SLM (selective laser melting) process. The coaxially aligned measurements of the melt pool morphology acquired during single-track processing have the capacity to spot process defects and proficiently ascertain the final track quality. DePond and his team [77] employed an SD-OCT-based detection system to gauge the surface roughness and layer-to-layer height details after the laser processing. Afterwards, a high-fidelity multi-physics simulation was introduced to elucidate the fundamental physics disclosed by the SD-OCT measurements and high-speed imaging. This provided valuable insights into the defect formation mechanism, which can be harnessed to optimize process parameters and enhance the print quality. With regard to Post-process OCT monitoring, future studies will concentrate on extracting the universal features based on OCT to correlate the surface topography information with welding quality, such as electrical resistance, penetration status, and defects under arbitrary conditions. Additionally, these studies will also consider the adaptation and integration of the OCT monitoring technique and the establishment of an OCT-based feedback control within an industrial environment.

4.4. All-process monitoring

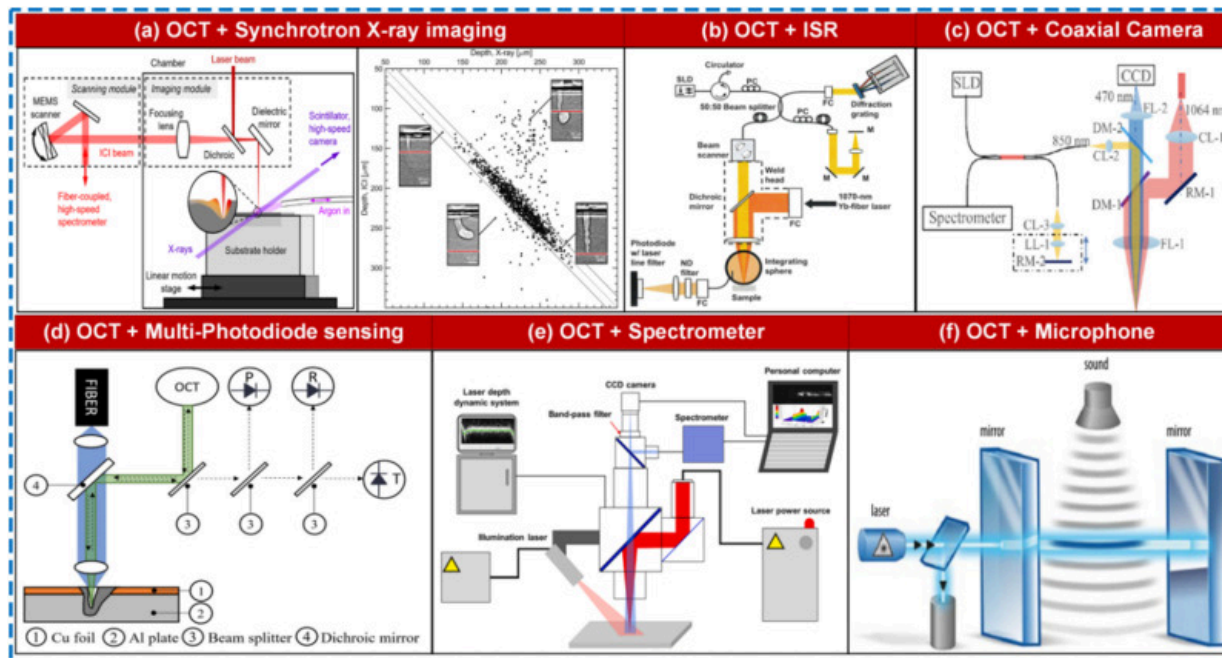
To achieve better *in-situ* detection and control throughout the entire laser welding process, only a few studies have delved into the applicability and feasibility of OCT as an All-process monitoring system. Dupriez et al. [30] developed an omnidirectional high-resolution OCT-based all-process monitoring system that encompasses functions such as seam tracking, quality identification, and quality assurance (QA). These functions are executed concurrently to ensure that weld quality standards are met. For the laser welding process of copper hairpin windings, Baader et al. [78] outlined the potentials and applications of all-process monitoring in the context of the deformed surface topography of the weld shape. With regard to Pre-process monitoring, it was demonstrated that OCT can detect certain deviations even before the welding process commences. Likewise, the post-process monitoring function allows for the assessment of the final weld shape. Additional experiments confirmed that the OCT data collected during the welding process can be utilized to calculate the keyhole depth and approximate the actual depth.

In addition, Fleming et al. [79] have applied the *in-situ* ICI system to track and control the evolution of the morphology in 3D powder-bed fusion. The height map obtained from Pre-process monitoring can be utilized to check the uniformity of the precursor material. Furthermore, the height map derived from Post-process monitoring not only shows the transformation of metal powder into solid metal but also discloses the removal of metal powder at the edges of the final layer. As a result, this research is capable of quickly detecting surface roughness, recoater blade damage, and powder packing density, thus making it easier to measure the quality of both the powder and solid layers. Recently, along with the development of the OCT sensing technique and the growing demand for industrial product quality, All-process OCT monitoring has demonstrated significant potential and commercial value in the fields of laser precision manufacturing. It can directly determine the location of the seam joint, the penetration depth, as well as the weld bead. This allows for the detection and eventual correction of manufacturing faults during the early stages of the processing. OCT holds the promise of optimizing the entire manufacturing process. Thanks to its capacity for conducting real-time, direct, and

independent measurements that remain unaffected by environmental elements and the processed laser light. It paves the way for contemporary online laser guidance while enabling simultaneous process monitoring. Moreover, it presents a simple yet innovative quality assurance solution for high-performance, high-throughput laser welding, all at a low cost per part.

5. Combination of OCT with sensing techniques

As a state-of-the-art interferometric imaging technique, OCT is presently being extensively utilized for the real-time monitoring of the morphology of keyholes and meltpools during high-power laser welding operations. Nevertheless, even under stable processing conditions, there are instances where OCT-based depth measurements deviate from the actual weld depths measured *ex-situ*, either showing shallower or deeper values. As expounded in Section 3.1, the origin of these outliers remains ambiguous, whether they stem from imaging artifacts, multiple scattering events within the keyhole, or practical variances arising from keyhole dynamics. This engenders a considerable conundrum in discerning the etiologies of OCT-data outliers and further impedes the precise extraction of weld depth. Fundamentally, OCT measurement data is efficacious only when a sufficiently profound capillary commences to form. For instance, in the conduction welding regime where the laser energy density is sub-critical, the absence of the keyhole mode would preclude the OCT sensor from accessing the actual penetration depth. Ergo, the sole utilization of the OCT system for the determination of diverse welding modes and seam depth is both inaccurate and insufficient. To surmount the aforementioned challenges, it is imperative to integrate other suitable sensing methodologies to effectuate a comprehensive analysis and inspection of the laser welding process. Such an approach is conducive to more efficacious monitoring and control of the welding process and quality, as illustrated in [Fig. 15](#).



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Fig. 15. OCT system with various sensing techniques: (a) synchrotron X-ray imaging [81]; (b) Integrating sphere radiometry [[95], [96], [97]]; (c) Coaxial camera imaging [100]; (d) Photodiode sensing [98]; (e) Spectrometer [101,102]; (f) Microphone [105].

5.1. Synchrotron X-ray imaging

High-speed synchrotron X-ray imaging serves as an operando external verification method for the OCT system when it comes to tracking keyhole dynamics under the interaction between laser and materials. By simultaneously capturing the complete two-dimensional keyhole profile, it can enhance our comprehension and handling of outliers present in OCT depth data, particularly in cases where there is a significant disparity between OCT and X-ray measurements. Moreover, the X-ray imaging technique is instrumental in analyzing the multiple reflections of the laser beam within the keyhole,

which has long remained an enigma in the OCT monitoring of the laser keyhole welding process.

Consequently, Fetzner et al. [80] explored the capillary dynamics through the utilization of time- and spatially-resolved inline X-ray imaging and OCT techniques. The OCT system enables considerably higher acquisition rates, reaching up to 70kHz, whereas the X-ray system operates at a rate of 1 kHz. These measurements are correlated with time-varying laser power, thereby providing profound insights into the dynamics of the laser welding process. In contrast to the aforementioned study, Fleming and his team [81] integrated synchrotron X-ray imaging (with a frequency of 140kHz) with high-speed OCT (operating at 280kHz and possessing a higher sampling frequency). Their aim was to decipher the intricate interplay between the laser processing beam and the OCT beam, as well as to uncover the influence of processing parameters on the OCT signals. This integration can further clarify the origins of OCT-data anomalies within the context of keyhole dynamic behavior. However, its resolution (at 100 μ m) is relatively limited considering the size of the keyhole in question.

5.2. Integrating sphere radiometry

According to Ref [82], the dynamic behavior of the vapor depression (keyhole) has the potential to significantly alter the number of laser-beam reflections. This alteration is inherently associated with the instantaneous energy coupled into the system. However, there is a conspicuous dearth of direct *in-situ*, experimental evidence regarding this rather nebulous relationship, thereby complicating the understanding and control of laser-based manufacturing processes. Recent remarkable advancements in X-ray imaging have managed to capture the melt-pool fluid flow as well as the two-dimensional profiles of vapor-depression formation and evolution.

Nevertheless, this approach is evidently constrained by the stringent measurement requirements and the limited availability of synchrotron light sources, as documented in Refs [[83], [84], [85], [86], [87], [88]]. In recent efforts to understand the energy coupling

into a metal, integrating sphere radiometry (ISR) has been employed to measure the absolute absorptance of laser light illumination, as seen in Refs [[89], [90], [91], [92], [93], [94]]. Against this backdrop, Allen et al. [[95], [96], [97]] were the first to combine two simultaneous state-of-the-art real-time OCT and ISR techniques with the aim of exploring and verifying the positive correlation between the highly dynamic keyhole morphology and laser energy absorptance. This work is capable of providing unique and profound insights into the underlying physics of the laser processing, which is beneficial for deterministic modeling and inline quality assurance.

5.3. Optical sensing

Unlike the above-mentioned X-ray and ISR techniques, the optical radiation emitted from the molten pool/plume constitutes a cost-efficient and efficacious sensing modality. Stemming from the localized interaction zone between the laser beam and the metallic material, the optical signals are distributed across multi-waveband emissions: In the spectral range of 400–850nm, corresponding to Ultraviolet (UV) and Visible (VIS) wavelengths. Notably, the UV/VIS radiation predominantly emanates from atomic transitions and bremsstrahlung within the plasma plume. In the region of 1000–1100nm, which is associated with laser reflection. In the 1100–1800nm band, attributed to Near-Infrared (NIR), where the IR radiation is emitted from the hot molten pool. Based on the diverse principles of optical signal acquisition, the principal optical sensors encompass photodiodes, cameras, spectrometers, and microphones. Subsequently, we will deliberate upon the combination of OCT and sundry optical techniques for the surveillance of the LKW process.

5.3.1. Multi-photodiode sensing

The low-cost and high-speed acquisition photodiode sensor finds common application in the monitoring of laser welding. This is due to its capacity to track crucial welding characteristics, such as keyhole and melt pool behaviors, plasma fluctuations, and spatter formation. However, the output of a low-resolution photodiode sensor typically yields a

unitless magnitude indicative of relative spectral response. This output is sensitive to multiple factors, including the distance between the sensor and the measurement site, the base materials employed, and the process parameters in effect. Consequently, the photodiode sensor necessitates the collection of extensive and effective experimental data, along with the establishment of classification criteria specific to each laser processing condition. Nevertheless, this approach is confined to discrete and qualitative quality evaluations.

To tackle these issues, Brežan et al. [98] devised a composite detection methodology by integrating multi-photodiodes with an OCT system. The objective was to measure the weld penetration depth during Cu-to-Al remote laser welding. Their research entailed analyzing and discussing the synergistic effects and competitive advantages of both sensors. This was accomplished by varying the laser power and focal offset within an industrial environment. Moreover, Beck et al. [99] posited that relying solely on the OCT depth has been demonstrated to be inadequate for accurately discerning different welding modes. As a result, they recommended the combined utilization of the OCT system with an InGaAs-photodiode to quantitatively estimate the welding mode during pure copper welding with a green laser source.

5.3.2. Coaxial camera imaging

In-process monitoring within the X-Y plane, which focuses on the surface features of the molten pool, represents a well-established research area. In this context, high-speed CCD cameras are commonly employed to directly image the molten pool and keyhole. These technological approaches are capable of providing critical surface information, such as the surface temperature distribution, molten pool geometry, particle splashing, and the characteristics of the metal vapor plume. The monitoring system in question adopts a synchronized high-speed coaxial observation method and integrates it with deep learning models. Although the high-speed CCD camera is effective in measuring the size of the molten pool, which exhibits a strong correlation with the contact spot area, it encounters limitations when it comes to predicting the depth of penetration.

Consequently, Li et al. [100] developed a setup that combines an OCT system with a coaxial high-speed camera. The purpose of this combination is to synchronously capture the three-dimensional information of the molten pool. Subsequently, a multimodal deep learning network was constructed to fuse the data sourced from these two modalities, aiming to predict the depth and contact width of the molten pool. Experiments were carried out under different laser power and welding speed settings. Additionally, the cross-sections of the weldment were cut and subjected to micro-imaging, which served as the ground truth for validation. In comparison to the individual sensing systems, the proposed fused system demonstrates enhanced precision in predicting both the depth and the contact width of the molten pool.

5.3.3. Spectrometer sensing

Optical spectrometers are frequently utilized to analyze the spectroscopic behavior of light. This is achieved by dispersing the light into its constituent wavelengths through the use of a prism or diffraction grating, and subsequently measuring the emission intensity at each individual wavelength with the aid of a photodiode detector or a charge-coupled device (CCD). In comparison to the 1D time-series data obtained from photodiodes, spectrometers are capable of furnishing more precise information for the analysis and calculation of keyhole plasma characteristics during the laser welding process. Kang et al. [101,102] investigated the applicability of a low-cost spectrometer as a depth estimation sensor for laser welding. A spectrometer with a sampling rate of 100Hz was chosen as the sole sensor, and a deep learning model was developed to predict the penetration depth in the laser welding of steel plates as well as Al/Cu overlap joints. The prediction performance of the proposed model was further validated by the welding depth data collected from the OCT measurement system.

5.3.4. Optical microphone

Apart from the optical monitoring systems based on photodiodes or spectrometers, the application of non-contacting microphones for detecting acoustic emissions during

manufacturing processes has demonstrated promising outcomes in the manufacturing industry in recent times. When conventional capacitive microphones are utilized, these systems face challenges due to interference from surrounding industrial noise. This is because the measuring range of such microphones is relatively narrow, typically being limited to a 100kHz bandwidth. Experiments have indicated that the optical microphone [103,104], which operates on the principle of interferometry, consists of a control unit equipped with a laser. This laser emits light that travels through an optical fiber to a sensor head. The optical microphone is capable of acquiring high-frequency airborne ultrasound signals (exceeding 100kHz), which can potentially be employed for monitoring keyhole welding. Consequently, Authier et al. [105] emphasized the results obtained from the recent combination of the ultrasound-sensitive membrane-free optical microphone and OCT system. By jointly interpreting the ultrasound spectrograms and the depth curves of the keyhole, a better understanding of the process dynamics can be achieved, and the determination of ideal welding parameters becomes possible. In particular, this enables the adjustment of the temporal modulation of the laser source power. Although this research is still in its nascent stage, the combination of OCT with an air-coupled ultrasound optical microphone appears to be a promising approach for the establishment of future inline control processes in laser welding.

In conclusion, these meticulously obtained observations hold significant value in propelling the development of intricate models and devising efficacious active control strategies. They unequivocally underscore the crucial role played by hybrid monitoring tools, which possess the remarkable capacity to concurrently gauge multiple dimensions of the process. The copious and synchronized nature of this information empowers researchers and engineers with a more comprehensive and profound comprehension of the LKW process. This, in turn, paves the way for the implementation of precise control measures and optimization procedures, culminating in a remarkable enhancement of the overall performance and quality standards within the realm of laser welding applications.

6. Combination of OCT with machine learning

In general, evaluating temporally high-resolution data presents unique challenges, particularly due to the complex and dynamic phenomena occurring in the process zone that can affect the estimation of weld seam quality. Machine learning (ML) techniques offer a promising solution, as they are capable of self-learning and extracting valuable insights from original data. These ML-assisted techniques have been widely applied in tasks such as nonlinear approximation, prediction, and classification within dynamic and intricate industrial processes [[130], [131], [132]]. Consequently, ML has emerged as a straightforward yet powerful tool for analyzing OCT-recorded depth data and interpreting this information in relation to welded seam properties. Notably, ML facilitates the implicit utilization of information contained in the OCT signal, which pertains to process-determining effects and phenomena that influence quality characteristics. Table 3 provides a summary of the integration of OCT with various ML techniques for the quantitative evaluation of weld quality: surface quality and internal quality.

6.1. Surface quality predicting

Firstly, a ML approach can be employed to uncover the correlations between the weld depth signal and the weld surface quality. This highlights the significant information contained in the OCT signals regarding the process phenomena that influence surface quality. To achieve this, a pre-processing of the original OCT data and a thorough analysis of the topographic information were conducted by Ref. [63]. The processed data were then utilized to train an artificial neural network (ANN) model to predict surface quality, distinguishing between good and poor outcomes with commendable accuracy. Recently, the OCT technique has become the preferred choice for quality monitoring in laser beam welding of hairpins. In this context, Raffin, et al. [106] employed ML and deep learning (DL) methods to assess weld defects by analyzing post-process OCT data. They also compared the accuracy and effectiveness of traditional ML techniques with contemporary DL methodologies in defect detection, offering a promising alternative to 2D-camera sensing in hairpin production.

In their research on laser deep penetration welding of concealed T-joints (web-face sheet), Pordzik and colleagues [107] repositioned the OCT probe beam in alignment with the laser beam to effectively capture the asymmetric keyhole features that correspond to deviations in the weld path. They subsequently employed analytical methods alongside an ANN model to quantitatively assess the OCT signals, aiming to enhance the prediction accuracy of the weld path deviation direction. The experimental findings further validated that the integration of OCT with ML techniques is proficient in distinguishing the directions of lateral path deviations, thereby optimizing the control of the weld path in hidden T-joints.

6.2. Internal quality predicting

Despite the aforementioned predictions regarding weld surface quality and path deviations, a significant challenge remains: integrating OCT with ML techniques to *in-situ* assess internal quality factors such as keyhole stability (KS), depth of penetration (DOP), seam width at the interface (SWI), and porosity. The application of ML methods to characterize the transient behavior of weld depth is particularly promising. For instance, Schmoeller et al. [108] introduced a generative Beta-Variational Autoencoder (β -VAE) model to predict laser weld depth across a wide range of processing conditions. This data-driven ML approach effectively captures the complex physical phenomena affecting weld depth, laying the groundwork for experience-based process control facilitated by OCT depth measurements. Over the years, Shu et al. [51] developed a long short-term memory (LSTM) network aimed at capturing the historical OCT-based information of keyhole depth to predict penetration depth at a specific moment. Experimental results demonstrated that the prediction model achieved high accuracy and robust generalization capabilities, enabling effective monitoring of penetration depth during the laser welding process. In addition, Li et al. [109] introduced an innovative real-time monitoring method for aluminum-copper dissimilar metal during adjust-ring-mode (ARM) laser welding. This method leverages OCT-based depth signals alongside plasma plume spectral signals. Advanced signal processing techniques were utilized to denoise and analyze both OCT and spectral signals, demonstrating a strong correlation between

multiple signals and key features. Ultimately, a BPNN-based model was developed for SWI prediction based on these multi-signal diagnostics. The findings indicate that this approach achieves an impressive average error of merely $10.7\mu\text{m}$, ensuring high-precision predictions of SWI in ARM laser welding.

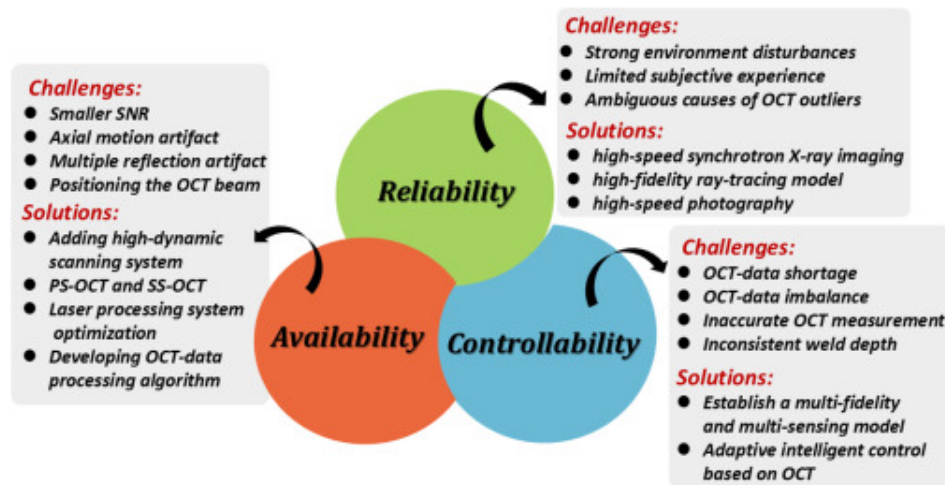
In parallel, to gain a deeper insight into the relationship between OCT signals and local porosity, Ma et al. [67,68,110] conducted data reconstruction pertaining to the penetration depth of aluminum alloy laser welding, as captured through OCT. Utilizing the 3D morphological characteristics of the keyhole obtained from multi-sensing signals, they developed several machine learning models, including ANN, Deep Belief Networks (DBN), and Sparse Autoencoders (SAE). These models were designed to enable real-time identification of keyhole-induced porosity during the laser welding process, eliminating the need for post-weld offline destructive inspections [113,114]. Future research will concentrate on the classification of weld seam quality using objective criteria, which significantly impacts predictive accuracy. Additionally, a comprehensive examination of the optimization of ML algorithm architectures will be undertaken. This approach aims to enhance the effectiveness of these algorithms in accurately assessing weld quality and improving overall performance in predictive modeling based on OCT measurement.

7. Major challenges and potential solutions

Despite remarkable advancements in the OCT-based all-process monitoring and inline quality assurance of the LKW process, in light of the intricate characteristics manifested in the laser-metal interaction, there still exist several significant challenges that urgently need to be addressed. These challenges and solutions can generally be categorized into three crucial aspects, as illustrated in Fig. 16.

- **Reliability:** centering around the precision and credibility of the data and results output by the monitoring and assurance systems;

- **Availability:** emphasizing the ease of use, adaptability, and compatibility with diverse working conditions of the relevant technologies and means in the actual production process;
- **Controllability:** focusing on how to precisely regulate various parameters using OCT to real-time ensure the stability of the process and the consistency of product quality.



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Fig. 16. Major challenges and potential solutions for *in-situ* monitoring the LKW process.

7.1. Reliability

7.1.1. Challenge 1

Intuitively, by capitalizing on the optical path variation, the OCT probe beam is capable of directly penetrating into the profound laser-material interaction zone within the metal workpiece, affording us a unique window to peer into the actual occurrences during the

quasi-stable LKW process. Nevertheless, the OCT signals obtained are highly susceptible to a multitude of potent interferences e.g. plasma plume, spatters, and thermal radiation [48]. Even under meticulously controlled and ostensibly stable processing conditions, discrepancies frequently emerge when comparing the OCT depth measurements derived from the keyhole that vapor cavity meticulously formed at the laser beam spot with the *ex-situ* measurements of the weld depth. The OCT measurements might appear anomalously shallow, potentially due to the occlusion of the beam by spatters or the attenuation caused by multiple reflections within the vapor capillary. Conversely, they could also seem deceptively deep, perhaps as a result of signal amplification from certain reflection patterns or the misinterpretation of echoes.

Simultaneously, in the current research, the *modus operandi* for interpreting and handling OCT data predominantly hinges on subjective empirical knowledge [123]. This is largely attributable to the nascent stage of comprehending the full complexity of OCT signal behavior. The origin of OCT outliers, those data points that deviate conspicuously from the norm, remains shrouded in mystery. Specifically, it remains uncertain whether these outliers stem from imaging artifacts or the intricate scattering/absorption effects. As the demand for precision in LKW processes rises, overcoming these obstacles becomes not only a scientific necessity but also a practical imperative for advancing manufacturing technologies.

7.1.2. Solution 1

To address the aforementioned challenges effectively, the most advisable solution at present is to utilize high-speed synchrotron X-ray imaging technology. This enables *in-situ* tracking of the intricate dynamic alterations within the interior keyhole and a detailed analysis of the multiple reflection procedures taking place inside the keyhole during the laser-metal interaction [[124], [125], [126]]. It can offer an unprecedented level of precision and temporal resolution, allowing researchers to capture even the most fleeting changes in the keyhole structure and the behaviors of the reflected beams. Alternatively, there exists a more cost-effective approach. By combining the high-speed

photography technique with a high-fidelity ray-tracing model, we could directly monitor the behaviors of the surface melt pool and the keyhole. This combination not only provides a clear visual representation of the physical phenomena but also facilitates a comprehensive description of the laser beam's propagation path and its interactions within the vapor capillary. The high-speed photography freezes the rapid changes in the melt pool and keyhole, while the ray-tracing model reconstructs the underlying optical pathways based on physical principles, offering complementary insights [127,128].

Through a comprehensive and detailed observation as well as simulation of the entire evolution process of the dynamic keyhole, from its initial formation to its subsequent development and eventual stabilization or dissipation, we can substantially enhance our comprehension and management of outliers in OCT data. This holistic approach will further unveil the concealed physical principles underlying OCT measurement and high-speed X-ray imaging, thereby providing valuable clues for a more refined interpretation of capillary depth measurements. Looking ahead, future research endeavors must focus on cultivating a profound understanding of the relationship between the propagation characteristics of the measuring beam and the morphological features of the keyhole. This will undoubtedly lay a solid foundation for optimizing the laser welding process and ensuring the production of high-quality products.

7.2. Availability

7.2.1. Challenge 2

In the LKW process, both in labs and industrial settings, obvious oscillations in the keyhole root depth are commonly observed. Coupled with the difficulty in reaching the keyhole bottom due to complex fluid flow and steep thermal gradients in the molten pool, many complex problems arise. Among them, the axial motion artifact is the main cause of significant losses in OCT signals. The signal losses inside the keyhole are irregular, constantly troubling data acquisition. This leads to a large number of outliers that deviate from the true values obtained theoretically or experimentally, reducing the

SNR of OCT measurements [42]. It affects measurement accuracy and data set reliability, posing difficulties for welding process control and optimization decisions. Particular in dealing with high-reflectivity metals like aluminum and copper, the OCT beam retains a lot of energy after multiple reflections off the keyhole wall. Once captured by the system, it generates artifacts based on multiple reflections, which becoming extremely difficult to analyze complex OCT data and estimate the keyhole depth/width with high precision. Even after undergoing multiple reflections, which are exacerbated by the excellent reflective properties of these metals, the OCT beam can still retain a significant amount of which is ejected molten particles, both act as formidable sources of interference. They cause the OCT beam to scatter in unpredictable directions and generate additional interference signals, effectively drowning out the true signal and making it extremely arduous to sift through the complicated OCT data. This complexity extends to the estimation of essential physical parameters, such as the keyhole depth, where achieving a higher measurement accuracy becomes a seemingly insurmountable hurdle [45].

In dynamic welding scenarios, where the processing laser beam follows a nonlinear path and the weld direction keeps changing, the keyhole usually lags 0–0.4mm behind the laser beam [129]. Finding the ideal measurement position of the OCT beam relative to the keyhole bottom in such a complex and changing environment is very challenging. Given the continuous relative motion of the laser beam, the research findings apply to galvanometer-based processing. However, achieving precise position synchronization in galvanometer measurement remains a key challenge and a focus for future research to enhance OCT measurement capabilities and availability for more efficient and high-quality manufacturing. To address the aforementioned challenges, potential solutions can be explored from the following two aspects:

7.2.2. Solution 2

7.2.2.1. Optimizing the OCT and laser processing systems

Firstly, focusing on the OCT measurement system, by integrating a high-precision and high-dynamic scanning system that can superimpose the welding motion, it becomes possible to achieve two significant objectives. On one hand, it enables the determination of the optimal measurement position of the OCT beam relative to the processing laser point. This precision positioning is crucial as it directly impacts the accuracy of the data collected. On the other hand, it empowers the system to track any welding trajectory, whether it is a complex, curvilinear path or an irregular oscillation geometry that might occur during the welding process. Consequently, a reliable calibration method for the measurement system is indispensable. This method should be designed to accurately position OCT beam across the entire processing field of the scanning optics, guaranteeing that the deepest keyhole can be precisely measured. With the continuous advancement of optical polarization and swept source-related technologies, the development of novel polarization sensitive OCT (PS-OCT) [117,118] and swept source-OCT (SS-OCT) [119] systems holds great promise. These innovative systems have the potential to mitigate the adverse effects caused by multiple reflected beams on the keyhole depth measurement. By doing so, not only can the accuracy of the depth measurement be enhanced, but also a higher imaging sensitivity and superior imaging quality can be achieved. The optimized OCT measurement systems are expected to substantially improve the SNR and the final measurement accuracy in future applications.

Secondly, optimizing the laser processing system to stabilize the melt pool and the keyhole is of paramount importance. This involves several key steps. Selecting appropriate laser sources is a fundamental consideration. For instance, adjusting ring-mode (ARM) laser, hybrid lasers such as semiconductor-fiber or blue semiconductor-fiber lasers, can offer distinct advantages [[120], [121], [122]]. These laser sources can provide better control over the energy distribution and penetration depth, which in turn facilitates the entry of most OCT beams into the bottom of the keyhole. Additionally, customizing the laser head according to specific welding requirements is another crucial aspect. By tailoring the laser head design, parameters like beam divergence, focus position, and spot size can be optimized to ensure that a significant quantity of OCT

beams can reach the keyhole bottom, thereby obtaining substantial amounts of valid signals. This will lay a solid foundation for accurate measurement and efficient welding processes.

7.2.2.2. Developing suitable OCT-data processing algorithms

Upon acquiring a vast amount of OCT data, devising a suitable processing algorithm emerges as the second crucial aspect. Although several classical statistical analysis and signal processing algorithms for extracting the melt depth have been put forward, there still exists ample room for augmenting the efficiency, accuracy, and applicability of OCT-data processing. A prospective solution lies in formulating a more universal, comprehensible, and actionable approach that does not necessitate parameter adjustment based on practical experience and extensive datasets. In recent years, treating the keyhole as a stochastic oscillatory system poses an intriguing physical conundrum and has become an open research topic. Researchers frequently place significant emphasis on conducting Fourier analysis of the signal oscillations originating from the keyhole and the melt pool. Deviations in the frequency components of the keyhole emission spectra can serve as telltale signs of melt ejections, defects, or other perturbations, as documented in references [115, 116]. Consequently, by integrating the keyhole dynamics mechanism with advanced time-frequency analysis techniques *e.g.* Wavelet Transform (WT) [134], Huang Transform (HHT) [135], or Empirical Mode Decomposition (EMD) [136], it becomes possible to extract more prominent physical attributes of the OCT signals while simultaneously reducing the data volume by several orders of magnitude, all without sacrificing the spatio-temporal resolution. These sophisticated OCT data processing algorithms are capable of accurately extracting the keyhole depth, filtering out specific imaging artifacts, and providing a rational interpretation of the raw OCT signals. As a result, in the near future, they hold great potential for seamless integration into existing and forthcoming commercial OCT applications, thereby enhancing the overall performance and reliability of OCT-based systems in various fields.

7.3. Controllability

7.3.1. Challenge 3

Due to the extraordinary complexity of the laser-material interaction, it is exceedingly challenging to construct a suitable physical model for quality assurance and process control. In recent years, combining OCT measurement with ML and adaptive control strategies has emerged as a highly promising solution, attracting significant attention. Nevertheless, several hurdles remain in the development of ML and control algorithms aimed at establishing an accurate “process-signature-quality” relationship model and further bolstering the stability of the LKW process. Firstly, acquiring sufficient and effective OCT-data related to various welding qualities, especially those associated with defective welds, typically demands time-consuming post-weld destructive inspections or costly radioactive detection. As a result, the prominent issues of OCT data scarcity and imbalance not only impair the predictive performance of the model but also undermine its controllability. Notably, OCT measurement data is only valid when a sufficiently deep keyhole is formed. In other words, relying solely on the OCT system to identify different welding modes and seam depths in the absence of the keyhole mode is both inaccurate and insufficient. This is because the absence of a keyhole means the OCT beam may not penetrate deep enough into the material to capture crucial details, leading to unreliable measurements.

Secondly, achieving precise in-line control of weld depth during the LKW process has been a focal point of attention for years. However, measuring and controlling a stable depth under high-speed welding conditions and with small focal diameters, particularly when the welding trajectory dynamically varies, presents another formidable challenge. Traditional Closed-Loop Control (CLC) solutions, which attempt to address this, are deterministic and rule-based [49]. They struggle to adapt to the stochastic environment characteristic of the highly dynamic LKW process, where rapid changes in temperature, material flow, and laser energy absorption occur constantly. Consequently, their success

has been rather limited. To surmount these challenges, potential solutions can be actualized from the following two aspects, which as follows:

7.3.2. Solution 3

7.3.2.1. Establishment of multi-fidelity and multi-sensing model

In response to the issues of OCT-data shortage and imbalance, one of the solutions is to establish an innovative multi-fidelity and multi-sensing model, which is increasingly emerging as the latest trend and research focus in the field of intelligent welding. Specifically, in this review, the term “multi-fidelity” refers to the three main types of fidelity data generated through the actual welding process, multi-physics simulation, and OCT *in-situ* measurement. By integrating these multi-fidelity data, a more comprehensive and high-quality database can be obtained based on the combination of physics-driven (process mechanism) and data-driven (machine learning) models. On the one hand, the physics-driven model can bolster the interpretability of the data-driven model. On the other hand, the outputs of the data-driven model serve as inputs to the data-driven model, which will furnish a richer set of OCT data and further enhance the generalization ability of the combined model.

To gain a better understanding of keyhole dynamics and determine the optimal process window, it is highly necessary to combine OCT with other suitable sensing techniques, such as X-ray imaging, ISR, or optical monitoring. By constructing a multi-sensing monitoring model that takes into account *a priori* information, it becomes possible to accurately identify different welding modes (heat conduction or deep penetration) and weld depth even in the absence of keyhole formation. However, currently, few studies have been dedicated to designing a multi-sensor monitoring system to comprehensively understand the “process-signature-quality” relationship. The scope and depth of multi-information fusion technology remain limited, and in-depth research is urgently required. In summary, establishing a multi-fidelity and multi-sensing model is of great value for

active quality control, highlighting the significance of hybrid monitoring tools that can simultaneously measure multiple aspects of the LKW process.

7.3.2.2. Adaptive intelligent control based on ML and OCT

To counteract the detrimental impacts on the stability of penetration depth caused by sporadic random interferences during the welding process, there is an urgent necessity to seek or explore a viable control strategy that can achieve precise control over the penetration depth. Adaptive intelligent control based on ML and OCT techniques represents a revolutionary approach in modern manufacturing and industrial processes. In recent years, with the exponential growth of data and the remarkable advancements in computational power, the fusion of ML algorithms and OCT technology has opened up new vistas for achieving highly precise and adaptable control. The real-time data acquired by OCT serves as a crucial input for the ML algorithms. As the OCT system detects minute changes in the keyhole geometry during laser welding, this information is immediately relayed to the ML model. The model then processes this new data, along with other relevant parameters, and determines whether any adjustments (laser power, travelling speed, *etc.*) to the welding process are necessary. The combination of ML and OCT enables a closed-loop adaptive intelligent control system, which could continuously monitor the process, detects deviations or anomalies, and make instantaneous corrective actions. In a manufacturing environment, this could mean maintaining a consistent weld penetration depth despite fluctuations in material properties or external disturbances. It could also involve automatically identifying and classifying defects in real-time, allowing for immediate remedial measures to be taken.

8. Conclusions

This work presents a comprehensive review of various aspects of OCT-enabled all-process monitoring and inline quality assurance in LKW. It presents the assorted classifications of *in-situ* surveillance, spanning the Pre-process, In-process, and Post-process phases, along with their pertinent applications. The overarching objective of this pursuit is to

substantially augment our foundational comprehension of the intricate nexus between process-signature-quality. Moreover, we embark on an exploration of the cutting-edge progressions in other state-of-the-art sensing modalities and machine learning (ML) techniques, delving into the potential pathways for their seamless integration into the OCT measurement framework. Subsequently, we conduct an in-depth deliberation on the principal challenges manifesting across three pivotal dimensions (reliability-availability-controllability) and proffer effective solutions. In totality, this all-encompassing review lays a solid foundation for more profound inquiries into the latent physics underpinning the LKW process, which will undeniably play a crucial role in expediting the development of deterministic models and enabling real-time and precise process control.

CRedit authorship contribution statement

Di Wu: Writing – original draft. **Peilei Zhang:** Methodology. **Haichuan Shi:** Data curation. **Qinghua Lu:** Investigation. **Zhishui Yu:** Supervision. **Yuze Huang:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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